
CAMPUSPHERE: ROLE-BASED ONLINE PORTAL WITH PROGRESSIVE WEB APP FOR CAMPUS MAPPING

A Capstone Project
presented to the Faculty of
College of Computer Studies
Camarines Sur Polytechnic Colleges

In Partial Fulfillment of the Requirements
for the degree Bachelor of Science in Information Technology

By

Carl Andrie D. Baldoza
Carl Ryan P. Lagrimas
Ocean Derek L. Regondola
Aaron V. Lasprillas
John Lawrence A. Paga

April, 2026

CHAPTER 1

INTRODUCTION

This chapter presents the justification, background, and purpose of the proposed system, CampusSphere, a role-based Progressive Web Application (PWA) campus mapping and information portal of Camarines Sur Polytechnic Colleges (CSPC). It describes the physical growth of the campus, lack of an official interactive campus map, and the disconnecting of spatial information in informal networks, as some of the factors that make navigation challenging among the students, the faculty, visitors, and other stakeholders. This chapter also places the project in the context of the larger trend towards smart campuses and smart universities, in which technology can be used to enhance the learning experience, user experience, and institutional efficiency [25, 27].

1.1 Project Context

Camarines Sur Polytechnic Colleges (CSPC) is a state higher education institution in the Bicol Region that has undergraduate and graduate programs in various subjects. The central campus has expanded to comprise various academic buildings, administrative offices, laboratories, and student services that are located over a comparatively large territory. New students usually come from various regions of Camarines Sur and other neighboring provinces, where most of them are not familiar with the campus format. Consequently, freshmen, transferees, and visitors on campus may find it difficult to move around the campus, especially during the first weeks of every semester.

At present, CSPC has been using static signage, the occasional use of printed or improvised campus maps, as well as informal directions from guards, staff, or fellow students to help people locate classrooms, laboratories, and offices. These practices have some support, but they are usually limited. Signages might not be covered on every building or every room; printed maps can be easily outdated when room assignments shift, and verbal instructions will rely on the presence and local knowledge of the individual staff or students. Similar situations have been documented in other institutions because of a repetition of confusion and delays

in locating buildings and rooms in an expanding college campus despite the existence of signboards and manual services [1]. Similarly, a project proposed by Takagi and Ogasawara (2025) developed a current and unified pedestrian navigation system for students and visitors that was integrated with a centralised spatial database that was accessible via smartphones, to address the lack of such a service for a complex campus environment [2]. The assessment of these systems suggests that, as sites grow, existing spatial supports are ineffective for cognitive navigation, efficient movement flow and positive user experience [1, 2, 4, 6].

In CSPC's case, there is currently no official, interactive digital map or campus navigation service that is available on the web or mobile platform at the moment. Spatial information such as buildings, offices, departments, and student services points is frequently distributed in many formats: bulletin boards, departmental or student-run social media sites, informal group discussions, and verbal directions. In recent studies on building 3D digital campuses, it has been found that this kind of fragmentation, in which spatial data and institutional information are spread in several systems rather than integrated into one official website, leads to confusion, redundant inquiries and use of invalid or inconsistent data [3, 25].

Empirical work on campus navigation also confirms the fact that new students and visitors are particularly prone to these difficulties. To improve cognitive navigation, digital systems in other universities were recommended following observations that freshmen and visitors were often late for class, missed appointments or required help to locate venues during orientation weeks [1, 2, 6]. While these studies employ different evaluation techniques, they demonstrate a common behaviour: in a large, complex sustainable campus, without appropriate cognitive navigation supports, users will often employ workarounds [1, 2, 6].

The connectivity conditions in the Philippine environment add another level of complexity to the navigation and information access situation. A mixed-methods study on the effectiveness of offline-capable mobile learning in low-connectivity areas found that 55% reported rare access to the internet, and only 15% reported regular access [13]. In the same study, it was found that 90%

reported that the unstable nature of the internet connectivity was difficult to manage when using online learning platforms, with the cost of data being a significant problem in terms of ongoing usage. [13] The study found that with the introduction of the offline-capable mobile learning application, the average test score increased from 48.6% in the pre-test to 72.3% in the post-test, an improvement of 48.8% [13]. This shows that applications designed to be used in low-connectivity environments can improve the learning outcome significantly, and it is essential to design the campus application with the capability to be used in low-connectivity situations.

Progressive Web Applications are particularly relevant in these contexts. PWAs leverage the ubiquity of web technologies with features of native applications such as installability, offline access via service workers, and responsiveness [11, 12]. The PWA system recorded an elevated technical performance. J-Lens showed an improvement of 26.8 percentage points in the average Lighthouse scores of 84.4% versus 57.6% in JGU Learning on performance, accessibility, best practices, SEO, and PWA features [26]. However, the PWA had a shorter fully loaded time (around 1.15 s vs. 1.5 s) and a faster DOMContentLoaded time (around 860 ms vs. 1.30 s) due to efficient caching and optimization of a single-page application [26]. even though it had to deal with more than twice as many HTTP requests. These findings build on previous results that PWAs are capable of providing more reliable and faster user experiences than existing web applications and can be installed as small-footprint apps on user devices [11, 12, 26].

On the strategic level, CSPC is also affected by the more general trend towards smart campuses and smart universities. According to Min-Allah and Alrashed (2022), the shift to a smart campus is as a result of incorporating emerging technologies to build an education ecosystem that meets contemporary learning needs [25]. In their model, smart digital platforms and applications are identified as essential services that are used to facilitate context-aware and user-friendly access to facilities on campus [25]. Raguro et al. (2020) suggest a Technology Management Framework of Smart University Systems that divides smartness criteria into five large criteria, such as Learning Environment and Educational Technology, and three pillars, namely, Technology Infrastructure,

Educational Pedagogy, and Government Collaboration, in the Philippine context [27]. The evaluation of Metro Manila universities reveals that the universities investing in the use of technology-supported learning environments and educational technology have more chances to achieve the level of smartness of a managed one, whereas the rest are still at the level of strategic plans but partially implemented [27].

Given CSPC's growing and spatially dispersed campus, the absence of a unified, official interactive map, and the fragmentation of location information on informal communication modes, the users experience the same navigation problems as were mentioned in other major academic institutions, where students and visitors are unable to find buildings and rooms without specific wayfinding devices [1, 2]. Documented connectivity constraints, which have been documented in similar educational settings, and the advantages of PWA-based and offline-enabled systems further point to the need for such campus applications to be functional even when network availability is intermittent or unreliable [11, 13, 26]. With respect to this, it is evident that there is a significant demand for a role-based, PWA-based campus mapping and information portal that is capable of helping visitors, new students, faculty, and administrators navigate the campus more effectively and access accurate spatial information from a single authoritative source. This system can also be used to help CSPC to alignment to newer developments of smart campus and smart university programs, which are concerned with context-aware, human-centered services and technology-enhanced learning environments [25, 27].

1.2 Purpose and Description of the Project

This project aims to design and develop CampusSphere, a role-based Progressive Web Application (PWA) campus mapping and information portal for Camarines Sur Polytechnic Colleges. The system is supposed to respond to the noted challenges of the visitors, new students, and even the faculty members who are facing the challenges of the growing and spatially dispersed campus of CSPC in the absence of an official interactive map and a single source of CSPC-wide spatial information. Through the use of web and PWA technologies, CampusSphere will offer an

affordable, high-quality, and user-friendly tool that will facilitate wayfinding and access to information related to the campus, both using desktop and mobile devices.

CampuSphere is envisioned as a PWA that can be installed and consists of an interactive campus map, a searchable building and building key location directory, and role-based student, faculty, and administrator dashboards. Users will be able to view building locations, get brief descriptions of the facilities, and search rooms or offices using keywords or categories. The system enables students and visitors to locate classrooms, laboratories, and service offices, and faculty and staff to easily confirm venue assignments and departmental locations. The administrative backend will allow authorised individuals to create and maintain data about the campus, like building entries, categories, and descriptions, and to maintain user accounts and roles, serving as a single point of truth for spatial information.

Since CampuSphere is designed to be a PWA, it would be able to provide offline support for some of its features, such as caching some of the essential data and assets on the client-side, so that users can still use the map and other data that has been previously loaded even if there is no internet connection or if the internet connection is slow. Studies about other educational systems developed using PWA-based and offline-capable applications have indicated that such applications can enhance their performance, responsiveness, and user experience, especially under low internet connectivity, compared to conventional web applications [11, 13, 26]. In addition, by consolidating campus spatial information into one unified portal, CampusSphere has the potential to help CSPC realize its long-term vision of becoming a smart campus/smart university, focusing on human-centered services enabled by technology, as discussed in environment [25, 27].

1.3 Objectives of the Project

The general objective of this project is to design and develop a CampuSphere: A Role-Based Online Portal with VR-Based Campus Mapping as a Progressive Web Application (PWA) to Camarines Sur Polytechnic Colleges (CSPC).

Specifically, this project aims to:

1. To identify the processes and required data needed in CampuSphere for:
 - 1.1. Campus building, office, and room information.
 - 1.2. Campus navigation; and
 - 1.3. Announcement dissemination;
2. To design and develop CampuSphere as a role-based PWA campus mapping portal by implementing the following system modules:
 - 2.1. VR Campus navigation module;
 - 2.2. Campus information module;
 - 2.3. Notification and Announcements module;
 - 2.4. User Management module;
 - 2.5. Installable PWA and offline access module;
3. To test the CampuSphere system using black-box in terms of:
 - 3.1. Functional testing;
 - 3.2. Usability testing;
 - 3.3. Security testing; and
 - 3.4. User satisfaction testing.
4. To deploy the CampuSphere system within the CSPC environment as a functional PWA platform for campus navigation and institutional information access.

1.4 Significance of the Study

The researchers believe that the development and implementation of CampuSphere: A Role-Based Online Portal with Progressive Web App for Campus Mapping would be beneficial to several stakeholders of Camarines Sur Polytechnic Colleges (CSPC), particularly in the following areas: improving campus navigation, improving access to information, and improving smart campus initiatives. The identified beneficiaries of this study are the students, particularly freshmen

and transferees, followed by other members of the academic community, the academic institution, and the information technology field.

Freshmen and Transferees Students. Most common problems are experienced by freshmen and transferees who are not yet familiar with the campus's floor plan. Without being provided with CSPC email credentials, CampusSphere provides freshmen and transferees with access to VR-supported navigation and information essentials that can help them get familiar with their environment before and during the first weeks of classes.

Students with Email. Students with official CSPC accounts will be provided with role-based dashboards that may include news, simulated enrolment-related information, assigned rooms, and direct links to navigation routes from their starting points to their classrooms and offices. They can also get familiar with the routes they need to take by following VR scenes and hotspots that simulate actual routes through the school's hallways and entrances. This can help reduce instances of lost students and those who are late for their classes, especially during peak academic periods.

Faculty Members. Faculty members can also benefit from CampusSphere by using it to look at their assigned rooms, room schedules, and announcements, and by using it to guide themselves through unfamiliar buildings or facilities. In cases where there are changes to the rooms or venues for make-up classes, seminars, and/or examinations, instructors can look at the portal for information rather than depending on instructions from other sources. This may help reduce navigational confusion and improve punctuality.

Camarines Sur Polytechnic Colleges (CSPC) as an Institution. For the entire CSPC, CampusSphere could be used in its digital transformation and smart campus development plans by offering a platform to students and others to explore and get information about the campus. A well-presented and visually appealing means of exploring the campus could not only improve the institution's efficiency by eliminating a lot of questions being asked and answered about where facilities are located but also boost its reputation as an efficient and user-friendly institution

System Administrators. This means system administrators will benefit from this system since they will have a central platform through which they can control users, VR scenes, navigation routes, room data, announcements, and simulated schedules. They will be able to encode room schedules, ensuring that users can view which rooms are being used at specific times using the details panel. This is significant since it will ensure that there are standardization, accuracy, and currency of data regarding rooms and building usage.

Guests, Parents, and Visitors. CampuSphere allows guests, parents, and other visitors to locate administrative offices, functional halls, clinics, and other important structures by selecting an origin place in the VR and a destination and then displaying the route in VR. This may help to improve visitor experience in the process of enrolment, orientations, graduations, and special events, as well as minimizing over-dependence on guards, staff, or students to give directions.

Information Technology (IT) Field. The project is of value to the IT field because it shows a practical use of the role-based web portal system, integrating the Progressive Web Application (PWA) and VR-guided navigation on a real campus. It demonstrates that modern web stacks, service workers, and immersive media can be applied to solve location-related issues and bandwidth limitations of resource-constrained institutions, and offers a practical case study and a possible reference design to developers and practitioners interested in smart campus and location-based services.

The Researchers. For the researchers themselves, the development of CampuSphere is an opportunity to gain additional technical experience in full-stack web development, PWA development, VR integration, database development, and system installation. The project also enhances their problem-solving, collaboration, documentation, and research skills, helping them develop professionally as IT practitioners in the future.

Future Researchers. CampuSphere may be considered by future researchers as a source of literature on the integration of role-based portals, PWA technologies, VR-based navigation, and the concept of smart campuses in the educational context, specifically in Philippine state colleges

and universities. The design choices of the system, constraints, and the assessment outcomes of this system can be used in future improvements or comparative research.

1.5 Scope and Limitations of the Project

This project includes the identification of the key processes and data required in campus navigation, information sharing, and building, office, and room data at Camarines Sur Polytechnic Colleges (CSPC); CampusSphere role-based campus mapping portal design and development based on Progressive Web Application (PWA) implementation; testing of its key modules; and the initial deployment of CampusSphere into the CSPC environment. CampusSphere system has user authentication and role management, role-based user dashboards, campus mapping, search and VR navigation, a panel for displaying details and institutional information, announcements (notifications) and room scheduling, administration and content management, and PWA installability and offline access. The project will also include black-box functional, usability, security, and user satisfaction testing, and the installation of the system on a suitable server at CSPC or an institutional platform, so it can be used as a tool for campus navigation and institutional information. Moreover, the spatial and mapping accuracy of the portal will be evaluated using similar software quality approaches to spatial digital twins using the ISO/IEC 25010 model [20].

The project has several limitations. CampusSphere does not use real-time GPS positioning, indoor positioning, or automatic rerouting based on a user's current location; the user can only follow preset start-to-destination paths modeled as VR scenes and interactive hotspots. All VR scenes, navigation paths, and room data are maintained and kept up to date by authorized administrators. The quality and quantity of data remain incomplete and incorrect unless it is maintained by an administrator when changes in campus layout, room assignments, or schedules occur. This release doesn't support integration with existing CSPC information systems, such as enrolment or student information; room schedules and announcements within the portal are mocked or manually coded to demonstrate its functionality and test it. Offline mode uses Service Workers and IndexedDB to store key MapLibre tiles, relational schedule data, and optimized VR panoramic

assets, reducing a need for real-time synchronization. The initial deployment uses a remote Supabase PostgreSQL database and a third-party delivery network for VR assets, but is highly available with a local institutional server deployment.

1.6 Project Dictionary

This section defines the key terms used in this study, clarifying both the overall meaning of the words and their particular use within CampusSphere used in Camarines Sur Polytechnic Colleges (CSPC).

CampusSphere. The suggested role-based online portal and Progressive Web Application to CSPC integrates VR-based campus navigation and institutional information services.

Destination. The destination where the user wants to get to, such as a specific building, office, or room. In this case, the destination is the last node of a navigation chain at which the pathfinding algorithm terminates, often defined as a Point of Interest (POI) or a target location [30]. In this study, the destination is used in CampusSphere as the final endpoint to determine the shortest and most accessible path from the user's current position.

Details Panel. The part of the interface that comes up when a room, office, or building hot spot is chosen that indicates the contextual details such as name, role, department, office hours, contact information, and plotting a schedule [28]. This study was used as a Details Panel of the UI which is activated in event of the choice of a particular room, office, or building hotspot. It offers access to immediate metadata like user position, department membership, office timetable, and contact data. In addition to displaying data, the panel is interactive, which means that a user will be able to plot and see schedules without the need to leave the main map interface.

Progressive Web Application (PWA). Design is made to be usable in a wide range of devices and network conditions. Its reactive nature allows it to be as functional as possible on other screen sizes, and the offline feature will also enable the user to remain interactive even without a reliable internet connection. This is because it extends the usability and scope of PWAs [11]. In this study, the PWA mentioned in the CampusSphere is installable on devices, whether phone or

laptop, but operates in the browser, and is capable of loading important maps and information of the campus, so that it would still be available when the internet is weak or not available.

Role-Based Access Control (RBAC). RBAC offers a hierarchical abstraction boundary between users and permissions allowing security administrators to operate access at scale by centering on roles and not user-permission relationships [14]. In this study, RBAC is used in CampusSphere to determine what features and information every type of user (student, instructor, administrator, guest) has access to, including dashboards, navigation tools, room or schedule manager.

Schedule Plotting. The representation of Times of usage of a room or facility, including class schedules or activity reservations, coded and stored by system administrators [29]. This paper defines Schedule Plotting digital form of representation of room or facility occupancy, which includes the timetable of classes and individual reservations of activities. These data are coded and stashed in a systematic manner by the administrators to guarantee real-time access points, in the accuracy of the entire campus map. The system enables users to view time-based occupancy and plan on space usage effectively in the portal by incorporating this in the interface.

Starting Point. The starting point of the path taken by the user to navigate, including the main entrance to the campus or the entrance of a specific building. In this case, the starting point is the first node in a wayfinding network, which can be defined as a position-sensing system (GPS or beacons) to determine the user location [32]. This is used in CampusSphere as the starting point when determining the shortest path taken to a given destination.

VR Scene / 360-degree View. An immersive truly digital world whereby a user is able to look around a space (360 degrees) using virtual reality technology [7]. In this study, VR scenes in CampusSphere are real campus areas like hallways, building exits, classrooms, and so forth, and corridors, enabling users to navigate predefined routes via interactive hotspots and directional elements.

Notes

- [1] Ajay Madhavan K. L. K., Hariharan K. S., Kowshigan M. V., Prasanna Kumar M. N., Sanjay T. S., Parthasarathy S., and Thangasankaran R. 2026. Campus Guidance System. *International Journal for Research in Applied Science & Engineering Technology (IJRASET)* 14, 1 (Jan. 2026), 1832–1840. <https://doi.org/10.22214/ijraset.2026.77223>
- [2] Takagi, Taro, and Goro Ogasawara. 2025. A Map Information Collection Tool for a Pedestrian Navigation System Using Smartphone. *Information* 16, 7 (July 2025), 588. <https://doi.org/10.3390/info16070588>
- [3] Yang, Zhi, Ming Wang, Jian Li, and Chao Liu. 2025. Construction of a Real-Scene 3D Digital Campus Using a Multi-Source Data Fusion: A Case Study. *ISPRS International Journal of Geo-Information* 14, 1 (January 2025), 19. <https://doi.org/10.3390/ijgi14010019>
- [4] Prathamesh Bhusekar, Shivanand Manyar, Hrishita Thanekar, and Vidya Chitre. 2025. Cross-Platform Development of a University Campus Navigation System Using Kotlin Multiplatform. *International Journal of Innovative Technology and Exploring Engineering (IJITEE)* 14, 9 (Aug. 2025), 1–7. <https://doi.org/10.35940/ijitee.H1116.14090825>
- [5] Nivedita Vibhandik, Pranav Kawale, Mayuresh Sonar, Mrunal Divate, and Aditya Sonar. 2025. Hybrid PWA Navigation System for College Campuses. *International Journal of Engineering Technologies and Management Research* 12, 10 (Oct. 2025), 1–6. <https://doi.org/10.29121/ijetmr.v12.i10.2025.1683>
- [6] Abdelaal, Mahmoud, Ali Elshater, and Hany Afifi. 2025. Enhancing Cognitive Navigation for Sustainable Campus Environments: Insights from Path Network Evaluation. *Sustainability* 17, 2 (January 2025), 514. <https://doi.org/10.3390/su17020514>
- [7] Achmad Chiron and Rusdi Hamidan. 2025. UX Analysis of the Virtual Tour 360 Application at Universitas Dr. Soetomo Campus. *Journal of Applied Informatics and Computing (JAIC)* 9, 1 (Feb. 2025), 73–78.

-
- [8] Priyanka Pujari, Shashidhar Sharan, Sarika Patil, Vaibhav Kalungada, and Sanket Khot. 2025. Designing an Immersive Virtual Reality Campus Tour. *International Journal of Research Publication and Reviews* 6, 5 (May 2025), 14316–14319. <https://doi.org/10.55248/gengpi.6.0525.1955>
- [9] Lanto Ningrayati Amali, Muhammad Rifai Katili, and Alif Perdana Sugeha. 2024. Development of Virtual Tour Reality Using 360-Degree Panoramic Images and Leaflet JavaScript. *Indonesian Journal of Electrical Engineering and Computer Science* 35, 1 (July 2024), 655–664. <https://doi.org/10.11591/ijeecs.v35.i1.pp655-664>
- [10] Gadbail D. R., Pawan Kamble, Shantanu Ghate, Kuldeep Khade, Rohit Dongare, and Prathamesh Walsinge. 2025. A Full-Stack Educational Platform with MCQs, Notes, and Role-Based Dashboards. *International Journal of Innovative Science and Research Technology* 10, 10 (Oct. 2025), 1253–1256. <https://doi.org/10.38124/ijisrt/25oct690>
- [11] Bangar Raju Cherukuri. 2024. Progressive Web Apps (PWAs): Enhancing User Experience through Modern Web Development. *International Journal of Science and Research (IJSR)* 13, 10 (Oct. 2024), 1549–1560. <https://doi.org/10.21275/MS241022095359>
- [12] Manu S. Rao and Rekha B. S. 2024. Progressive Web Apps in Cross-Platform Development: A Comparative Analysis and Evaluation. *International Journal of Scientific Research in Engineering and Management (IJSREM)* 8, 6 (June 2024), 1–6. <https://doi.org/10.55041/IJSREM35595>
- [13] Ibrahim Umar and Kasim Abubakar Argungu. 2025. Offline Capable Mobile Learning Solutions for Low Connectivity Regions. *International Journal of Sustainable Social Science (IJSSS)* 3, 5 (2025), 337–354. <https://doi.org/10.59890/ijsss.v3i5.107>

-
- [14] Albert Ramadhan Manik, Dedy Kiswanto, Muhammad Budi Akbar, and Jogi Purba. 2025. Academic Portal with MFA (WhatsApp OTP via Fonnte), Role-Based Access Control, and Logging System for Network Monitoring. *Jurnal Ilmiah Sistem Informasi (JUISI)* 4, 3 (Nov. 2025), 674–687. <https://doi.org/10.51903/qwz6pt87>
- [15] Ramesh Lavhe, A. N. Kalal, Rasika Lokolkar, Shreyas Kulkarni, Siddhesh Dhande, and Vishal Sharma. 2025. Academic Management through a Web-Based Student Profile System. *International Journal for Research in Applied Science & Engineering Technology (IJRASET)* 13, 5 (May 2025), 7347–7354. <https://doi.org/10.22214/ijraset.2025.71890>
- [16] Rusnida Romli, Muhammad Afiq Aznan, Lip Zhong Xian, Nur Anis Alia Bakhorrudin, Fatin Nur Hazwani Mohd Wazir, and Amar Raaj Singh. 2021. AR@UNIMAP: A Development of Interactive Map Using Augmented Reality. In *5th International Conference on Electronic Design (ICED 2020)*. *Journal of Physics: Conference Series* 1755, 012052 (2021), 1–7. <https://doi.org/10.1088/1742-6596/1755/1/012052>
- [17] Abiodun Durojaye, Amin Kolahdooz, and Alireza Hajfathalian. 2025. Enhancing Virtual Reality Experiences in Architectural Visualization of an Academic Environment. *EAI Endorsed Transactions on AI and Robotics* 4, 1 (2025), e3, 1–15. <https://doi.org/10.4108/airo.8051>
- [18] R. G. Suresh Kumar, Akash L., Adhithan R., Suriya S., and Yogeshwaran P. 2025. V-Campus Experience. *International Journal for Research in Applied Science & Engineering Technology (IJRASET)* 13, 5 (May 2025), 2698–2704. <https://doi.org/10.22214/ijraset.2025.70635>
- [19] Devaldi Akbar Suryadi and Endang Sulistiyani. 2022. Evaluation of Information Quality Using ISO/IEC 25010:2011 (Case Research: Menu Harianku Application). *International Journal of Innovation in Enterprise System (IJIES)* 6, 2 (July 2022), 119–132. <https://doi.org/10.25124/ijies.v6i01.165>

-
- [20] Alqarni, Ahmed, Saeed A. Ghafar, and Mohammed A. A. Majid. 2025. Assessing Spatial Digital Twins for Oil and Gas Projects: An Informed Argument Approach Using ISO/IEC 25010 Model. *ISPRS International Journal of Geo-Information* 14, 8 (August 2025), 294. <https://doi.org/10.3390/ijgi14080294>
- [21] Yarshini Thamilarasan, Raja Rina Raja Ikram, Mashanum Osman, Lizawati Salahuddin, Wan Yaakob Wan Bujeri, and Kasturi Kanchymalay. 2023. Enhanced System Usability Scale Using the Software Quality Standard Approach. *Engineering, Technology & Applied Science Research* 13, 5 (2023), 11779–11784. <https://doi.org/10.48084/etasr.5971>
- [22] Mohammed Khaja Sirajuddin. 2024. Cloud Technologies Revolutionizing Global Education: A Technical Case Study. *International Journal of Scientific Research in Computer Science, Engineering and Information Technology* 10, 6 (Nov.–Dec. 2024), 2084–2092. <https://doi.org/10.32628/CSEIT2410612400>
- [23] Vasyl P. Oleksiuk and Olesia R. Oleksiuk. 2021. The Practice of Developing the Academic Cloud Using the Proxmox VE Platform. *Educational Technology Quarterly* 2021, 4 (2021), 605–616. <https://doi.org/10.55056/etq.36>
- [24] Ghazwan K. Ouda and Qahtan M. Yas. 2021. Design of Cloud Computing for Educational Centers Using Private Cloud Computing: A Case Study. In *ICMAICT 2020. Journal of Physics: Conference Series* 1804, 012119 (2021), 1–8. <https://doi.org/10.1088/1742-6596/1804/1/012119>
- [25] Min-Allah, Nasro, and Salih Alrashed. 2022. A Smart Campus Framework: Challenges and Opportunities for Education Based on the Sustainable Development Goals. *Sustainability* 14, 15 (August 2022), 9640. <https://doi.org/10.3390/su14159640>
- [26] Dian Nugraha, Febria Anjara, and Safira Faizah. 2022. Comparison of Web Based and PWA in Online Learning. In *5th FIRST T1 T2 2021 International Conference (FIRST-T1-T2 2021)*. Atlantis Highlights in Engineering 9 (2022), 201–205.

-
- [27] Ma. Corazon G. Fernando Raguro, Ace C. Lagman, and Ronaldo A. Juanatas. 2022. Technology Management Framework for Smart University System in the Philippines. In Proceedings of the 2021 9th International Conference on Information Technology: IoT and Smart City (ICIT 2021). ACM, New York, NY, USA, 372–380. <https://doi.org/10.1145/3512576.3512642>
- [28] Madhura S, Mandara H, Meghana C, and Mr. Neelakantappa. 2025. Smart campus navigation system. *International Journal For Multidisciplinary Research* 7, 1 (January 2025). <http://dx.doi.org/10.36948/ijfmr.2025.v07i01.34970>
- [29] Silvia Mirri, and Catia Prandi. 2022. Designing interfaces to display sensor data: A case study in the human-building interaction field targeting a university community. *Sensors* 22, 9 (April 2022), 3361. <http://dx.doi.org/10.3390/s22093361>
- [30] Devonnie Clyde G. Baniaga, Gartly R. Cortez, Iordanis Keith D. Jacinto, Miella Carmel F. Rabago, Neil Bryan V. Real, Rein Arwen D. Santos, Jazhmaigne Lizney Ea C. Taylan, Keziah Faye M. Arellano, Johnny R. Pagaduan, and Jhansen Rey M. Obispo. 2023. *Development and Testing of a Digital Map Guide of Ilocos Norte National High School Main Campus*. Ilocos Norte National High School, Laoag City, Philippines.
- [31] Rachel T. Alegado DIT T. Alegado, Carlos Miguel M. Guzman, and Marinella Josephine B. Manuel. Retrieved March 8, 2026 Retrieved from https://www.academia.edu/126337201/NEUST_Wayfinder_Redefining_Campus_Navigation_with_Advanced_Technology_and_User_Centric_Design
- [32] Zoe Charis Belenioti, and Christos Kypri. 2024. *A Cutting-Edge Exploration: A Literature Review on Optimizing Museum Experiences Through Digital Transformation and VR Tour Software with 3DVista*. European University of Cyprus, Nicosia, Cyprus. http://dx.doi.org/10.1007/978-3-031-87019-4_17

CHAPTER 2

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter presents a review of related literature, studies, articles, and systems that provide the theoretical and conceptual foundation for modern campus navigation and information systems, particularly in relation to the current trends in Progressive Web Applications (PWA), Virtual Reality (VR), role-based information portals, and the different testing approaches; it also provides background information that justifies the importance and relevance of such systems by discussing initiatives of different institutions and developers, the existing research gaps and key features that lead to the enhancement of campus navigation, accessibility and information services, specifically in the Camarines Sur Polytechnic Colleges (CSPC).

2.1 Related Literature and Systems

The thematic categories that follow provide an empirical, conceptual, and technical framework for evaluating the flexibility of navigation systems, virtual reality, PWA, and role-based information portals in the context of academic institutions. Each of the following subsections provides an overview of related studies and systems, their major features, limitations, and the impact of the discussed frameworks on campus navigation and information portals.

2.1.1 Campus Navigation Systems (Non-GPS and Web-Based Approaches)

The latest trend in campus wayfinding systems in higher educational campus deals with an old problems of locating buildings, offices and facilities on busy campuses environments. As campuses grow and become more complex, traditional wayfinding techniques such as signs, maps and verbal directions with digital methods. This section examines the current non-GPS and web-based campus navigation systems with respect to their features, advantages and drawbacks.

Web-based navigation systems are used in the few universities to provide structured guidance without relying to real time GPS tracking, which can be inaccurate in concrete structures. Madhavan et al. [1] developed a Campus Guidance System which is a digital map that

will help new students and visitors to navigate the campus who can easily get lost in the growing campus environment. They had searchable location directories to minimize the need for assistance. Similarly, Anusiuba et al. [2] implemented a digital campus orientation and navigation system that incorporates an interactive campus map, academic classrooms and event schedules, services, showing coordinated digital platforms are more effective and convenient than paper based campus maps or even verbal directions. Modern web-native technologies are now being favoured over traditional game-engine-based and multiplatform-native navigation applications for campus. New advances in vector web mapping highlight the effectiveness of MapLibre GL JS to render highly configurable, interactive campus maps natively, avoiding high memory consumption by engines like Unity [3]. The use of vector tiles in MapLibre GL JS ensures smooth panning and zooming of large campus areas without the need for users to install apps, bringing AR into the mix to help users navigate in indoor environments such as classrooms, labs and corridors. This research demonstrated using mobile navigation with AR significantly enhance user performance in a building where GPS is not available Nordin et al. [4] took this idea further by developing a web-based campus tour system with integrated mobile augmented reality (AR) and they showed that, when used together, web maps and AR layers enhances the ability of a student to navigate their way around a new environment. To support these client-side mapping libraries, effective server side solutions with Node.js and Express.js have served us well. Node.js is an event-driven, supports scalable API and dynamically handles real-time data for campus directories and user authentication [5]. This JavaScript stack ensures that students and visitors are able to access structured data on buildings and suggested routes via any device with predefined building models, fixed routes and node-based pathfinding to help users navigate and find points of interest without using real-time location data.

These studies shows that web-based and mobile navigation systems are alternatives to static maps and verbal instructions especially for people who are new to a campus. When you have building data and predefined routes it is easier to find classrooms, laboratories and offices

especially when the campus is really busy like during enrollment period. The systems reviewed mostly have two-dimensional user interfaces and do not allow the user to get a feel of the space and different types of information for different users.

These findings are important for Camarines Sur Polytechnic Colleges. For example, new students and visitors often confused to reach the Registrar's Office, while the route to the Cashier inside the administrative buildings is confusing. Moreover, during peak traffic times, like enrollment periods, the lack of digital route navigation from the entry points up to the Student Testing and Admission Building results in major bottlenecks in foot traffic, revealing a need for structured, non-GPS based digital navigation on campus.

2.1.2 Virtual Campus Tours and 360-Degree Navigation

Virtual campus tours and 360-degree navigation systems has gained popularity in higher education as spatial awareness enhancing and visualising features of campus environments. These use of virtual panoramas and interactive hotspots f or users to interactively navigate campus spaces virtually or in preparation for an actual campus visit. This subsection looks at existing solutions, focusing on their design features, user experience outcomes, and limitations.

Choiron and Hamidan [6] studied user interactions with a 360 degree virtual tour application at a university campus and found users enhanced spatial knowledge through immersive panoramic views. The research showed that users learn more about the campus through scene-based navigation, using markers and hotspot icons, than from static image-based navigation. The usage of immersive orientation tools allows students from remote Bicol region municipalities who plan to enroll at CSPC to establish a connection between their remote research and their need to visit the actual campus. Pujari et al. [7] demonstrated that virtual reality tour designs use scene transitions and viewpoint controls together with real-world visual indicators to enable users to navigate different campus areas. The study demonstrated that users' ability to transition between scenes depended on the visual anchor quality, which included both recognizable landmarks and labeled hotspots. On the technical implementation of current systems uses dedicat2.1ed WebGL

panorama viewers which include Pannellum.js as their primary solution because it provides native support for equirectangular image display without the high-performance costs associated with conventional 2D mapping tools such as Leaflet[8]. Pannellum.js enables users to navigate between different scenes through its interactive hotspots which provide access to neighboring viewpoints and building entry points and interior spaces. The system needs media delivery networks like Cloudinary because high-resolution 360-degree images require extensive bandwidth and the network helps to deliver visual content efficiently through optimized asset compression methods. The combination of Pannellum.js immersive rendering and Cloudinary media optimization enables users to explore campus panoramic routes without using excessive device memory or mobile data limits. The system uses hotspots as interactive gateways which provide access to nearby views and building entrances and rooms and information pop-ups.

The reviewed literature consistently demonstrates that 360-degree virtual tours and VR-based navigation systems significantly enhance students' spatial knowledge and orientation on university campuses. Unfortunately, most of these projects are oriented mainly to the users' initial step, familiarizing them with the area appeal, ignoring support in moving around, like start-to-end path finding or personalized content according to the users' role. These results are also applicable at CSPC, a place where visually directed navigation would be a great help since visitors often lack effective on-site directory maps.

In the context of CSPC, this type of immersive visualization will be a great help, especially for freshmen who have never been to the campus and for external visitors who have come to attend institutional events and who will appreciate the advance spatial orientation leading to specific campus facilities.

2.1.3 Progressive Web Application in Education and Information Systems

Progressive Web Applications (PWAs) have become a major alternative to traditional web and native application types where they provide a single development model allowing installability, offline access, and fast loading performance. Schools choose PWAs as a way to fix

accessibility issues and at the same time improve performance since these are difficulties that they encounter. In this article, we explore the research on developing and deploying PWAs for education and information systems.

Cherukuri [9] conducted a thorough review of the capabilities of PWA which showed that these applications make use of modern browser features in their implementation of service workers and caching methods to ensure that they are reliable under various network conditions. The study found that service workers function as a background system that enables users to access application content during network outages because it stores essential resources. Rao and Rekha [10] conducted a comparative analysis of PWAs versus native web applications and found that PWA-based systems consistently outperform traditional web applications in terms of load time and offline reliability. In educational system evaluations, Nugraha et al. [11] conducted a study which compared the performance of a PWA-based online learning platform to a traditional web-based platform and found that the PWA showed better performance results and higher accessibility scores. Vibhandik et al. [12] developed a Hybrid PWA Navigation System, which enables college campuses to use offline caching, and device-agnostic functionality to provide better support for students and faculty who depend on mobile data.

This is particularly applicable to CSPC, the student body depends on personal prepaid mobile data because they cannot access institutional Wi-Fi. PWAs help users save mobile data and phone storage space because they allow users to avoid downloading large native app store applications. The system enables students, faculty members and temporary visitors to access CampusSphere portal from web browsers because it eliminates the need for installation thus providing an affordable and efficient solution to the entire campus community.

2.1.4 Role-Based Access Control (RBAC) and Smart Campus Portals

Role-Based Access Control (RBAC) is a widely adopted access management model that assigns permissions to defined roles, enabling scalable administration of multi-user systems. In academic portal development, RBAC ensures that students, faculty members, and administrators

access only the features relevant to their roles. This subsection reviews RBAC implementation in academic and institutional web systems.

Manik et al. [13] discovered that RBAC functions as an effective method to strengthen system security while decreasing administrative work and maintaining access control in academic multi-user systems. The researchers implemented RBAC together with WhatsApp OTP multi-factor authentication and a network monitoring logging system to show that structured role hierarchies decrease unauthorized access attempts while enhancing audit traceability. Lavhe et al. [14] showed that RBAC systems work effectively in web-based student profile applications by demonstrating that role-based dashboards permit users to access academic schedules, student records and administrative datasets while protecting sensitive information from unauthorized access. Their implementation showed that well-structured role boundaries simplify system maintenance and reduce the occurrence of user errors. According to Romli et al. [15] educational portals which implement RBAC create more distinct operational boundaries which enhance overall system usability. The implementation of RBAC at CSPC will enable the navigation portal to provide different features which include administrative room management access and student access to personal schedule information.

The reviewed literature confirms that RBAC is an effective and well-supported mechanism for managing multi-stakeholder access in academic portals. The implementation of RBAC together with authentication systems and structured content management systems leads to enhanced security, decreased information overload, and improved user experience through better user interaction design. The principles described here apply to campus information systems which operate for multiple user groups who require different types of information and access rights. The existing system benefits from RBAC-based personalization which yet remains untested in campus navigation and mapping systems according to existing literature.

For CSPC, implementing RBAC will enable the system to deliver suitable navigation and information services to students, faculty members, administrators and visitors according to their

designated roles. The first-time guest who wants to find the Registrar's Office needs a customized user interface that displays only relevant information because he will encounter too much institutional data including general student body laboratory schedules.

2.1.5 Offline-Capable and Low-Connectivity Solutions

Offline-capable systems have gained considerable importance in regions where internet access is unstable, intermittent, or economically restrictive. In educational contexts, unstable internet access limits learning outcomes and the benefits of information systems. This subsection reviews literature on offline-capable mobile applications and low-connectivity systems, examining their design strategies, effectiveness, and impact on system access.

Umar and Argungu [16] conducted a mixed-methods study on offline-capable mobile learning platforms designed for low-connectivity regions and documented significant connectivity barriers faced by learners. Their study found that 55 percent of respondents rarely accessed the internet and only 15 percent reported regular connectivity. Furthermore, 90 percent identified unstable network conditions and high data costs as major obstacles to using online learning platforms. The introduction of an offline-capable mobile learning application produced a substantial improvement in learning outcomes, with average test scores rising from 48.6 percent in the pre-test to 72.3 percent in the post-test, an improvement of 48.8 percent. The offline application achieved these results through caching mechanisms, local storage and asynchronous synchronization which enabled users to access previously loaded materials and perform basic tasks without an active network connection. Vibhandik et al. [12] demonstrated that PWA-based navigation systems which use offline caching together with service workers can maintain their functionality for students who use mobile data or campus Wi-Fi that operates intermittently. The process of caching high-resolution 360-degree imagery brings multiple technical difficulties which affect both data storage requirements and data transmission needs. Braun et al. [17] created a dual-engine intelligent caching system which handles 360-degree mobile content through advanced movement prediction technology that optimizes storage of panoramic content. The

study shows that caching large 360-degree image files on mobile devices becomes possible through the implementation of spatial locality-aware replacement systems that control storage management. The CSPC navigation application depends on these strategies to maintain access to campus panoramic scenes in locations with weak cellular signal strength which includes the ground floors of concrete academic buildings.

This is highly relevant to the physical layout of CSPC, where the institutional Wi-Fi network frequently struggles to penetrate the deep concrete infrastructures of the academic and administrative buildings. The system provides continuous access to pre-loaded 360-degree spatial maps and routing data for students who experience sudden connectivity dropouts during their indoor facility navigation through the implementation of robust offline caching solutions which eliminate physical dead zones as a barrier to campus navigation.

The existing offline-capable systems demonstrate effective performance yet researchers have not published sufficient studies that focus on the difficulties of storing extensive multimedia content which includes 360-degree panoramic imagery. The assets demand extensive storage space while special compression methods must be used to achieve efficient local storage without performance loss. Research about mobile content delivery shows that organizations need to control their large image collections through lazy loading, image compression and selective caching methods to avoid excessive memory usage that decreases application speed. The implementation of offline 360-degree navigation systems requires organizations to achieve two objectives which involve providing immersive content and handling device storage limitations.

2.1.6 Cloud and Academic Cloud in Educational Systems

Cloud computing has become a foundational infrastructure component in modern educational systems, which a scalable, cost-effective and centralised alternative to more expensive on-premise server infrastructure. Educational institutions use academic cloud platforms to solve their operational challenges, such as resource management, access and control and system

availability. This section discusses the existing research on the implementation of cloud computing and the usage of educational cloud systems in educational institutions.

Sirajuddin [18] examined cloud technologies in global educational institutions and identified that cloud-based deployments allow universities to deploy applications, databases, and academic tools on virtualized infrastructure, and this helps improve availability and eliminates onsite maintenance. The research showed how educational institutions can use cloud infrastructure to match their computing needs with actual requirements which helps them to operate their systems that support large and diverse user communities. Educational institutions need their operational requirements to be solved through Backend-as-a-Service (BaaS) platforms which is better than hardware-intensive private clouds and virtual machine platforms such as Proxmox VE. The operational needs of educational institutions required Backend-as-a-Service (BaaS) platforms because they delivered better results than traditional hardware-based private clouds and virtual machines that included Proxmox VE. For instance, platforms that include Supabase deliver a highly expandable centralized cloud system which provides strong PostgreSQL database functionalities together with effortless authentication mapping and safe hosting solutions that eliminate the need for users to manage their own local servers [19]. This allows the universities to operate their educational systems, their online platforms and their research systems in an efficient manner. Their study found that academic cloud platforms support effective management of access, resource allocation, and infrastructure costs, making them well-suited for deploying multi-user institutional portals. Studies have shown that using Supabase as an open-source backend platform in educational portals improves system availability while providing secure role-based data synchronization in real time. This architecture provides reliable uptime and centralized data management while drastically reducing administrative overhead and infrastructure costs, making it well-suited for deploying multi-user institutional portals [20]. Educational centers need to enhance system performance and security and monitoring capabilities. The study demonstrated that private cloud architectures deliver dependable system uptime together with centralized data

management capabilities. The system operated according to the established institutional data governance policies.

The reviewed literature establishes that cloud and academic cloud platforms provide educational information systems with infrastructure that demonstrates both reliable performance and capacity to scale and maintain operation. The deployments provide two main advantages by decreasing on-premises management tasks while supporting multiple users at once and allowing central management of educational materials. The properties apply specifically to systems that operate through the entire campus and need continuous system uptime together with their capacity to manage multiple data types and their requirement for permanent functioning across the institution. For CSPC, being dependent on cloud infrastructure has proven indispensable in order to ensure that the CampusSphere portal is highly available and scalable during the peaks of its operation period. This can be seen clearly during the university enrollment week, where thousands of students are accessing the real-time schedules since not only does cloud computing help make the system more interactive, but it also prevents the institutional servers from being overloaded.

2.1.7 Smart Campus and Smart University Frameworks

The concepts of smart campus and smart university involve the incorporation of information technology solutions, services, and systems that can simplify the processes involved and make learning more efficient. The models provide guidelines for implementing technologies in higher education institutions, with emphasis on contextually relevant services, systems, and user interfaces. This subsection will review some key smart campus and smart university framework and their impactation on technology development within institutions.

Polin et al. [21] conducted a systematic literature review using the PRISMA approach to establish a conceptual framework for the smart campus, identifying four primary domains: society, economy, environment, and governance. Their review found that smart campuses are increasingly being seen as small cities themselves and used as a testbed for digital technology whilst undertaking conventional teaching, learning and research. They identified context-aware

digital services such as navigation and information tools as infrastructure elements needed in order to enhance higher education's institutional digital maturity. In the Philippine context, Raguro et al. [22] proposed a Technology Management Framework for Smart University Systems where institutional digital maturity was broken down into five key aspects including Learning Environment and Educational Technology based on three main pillars Technology Infrastructure, Educational Pedagogy, and Government Collaboration. Their assessment of universities located in Metro Manila showed that those institutions that were making efforts towards creating technology-based learning environments were able to reach a managed level of smartness within the institution. They have pointed out that navigation services, information systems accessibility, and digitization were key elements necessary to proceed towards higher levels of digital maturity at Philippine State Colleges and Universities.

Smart campus and smart university always consider navigation and information as critical services in technology-enabled learning environments. Literature from international sources as well as Philippine sources suggests that there is a need to develop context-aware IT systems to move the institution to a higher degree of smartness. This normative approach will help institutions in determining the functional scope of their information systems and its consistency with their vision regarding the development of an institutional digital transformation. For CSPC, the use of these navigation and role-based information systems forms the base step towards attaining the level of smartness desired by the institution. Through the use of this model, CSPC will convert their inefficient system where freshmen had to depend on verbal directions to find their way through Academic Buildings or students had to visit different social media just to get updated from school news into a centralized, context-aware smart campus environment.

2.1.8 Software Quality Assurance and Testing Methodologies

Ensuring the functional correctness, usability, security, and overall user satisfaction of software systems can only be assured through the use of quality assurance and testing techniques. When developing web-based and campus information systems, one of the popular techniques

used is black box testing where the testing of system behavior takes place without any need to look at the actual internal code, thus making it ideal for evaluating the user interface of the system by end-users. The following section analyzes previous studies on testing techniques employed in education and campus environments with respect to the functional and non-functional requirements of campus information systems.

For functional testing, Suryadi and Sulistiyani [23] assessed the quality of information presented in a mobile web application using the ISO/IEC 25010:2011 standard, which outlines quality characteristics such as functional suitability, performance efficiency, reliability, usability and security. The study used structured test case design based on the ISO/IEC 25010:2011 standard to verify that the system's features functioned as intended; the study showed that structured test case design and execution led to significantly improved detection of defects before deployment. They found that structured test case design and execution by professional testers led to improved system reliability, and that this approach can be applied to assess the functional components of role-based portals and PWA-based systems. For usability testing, Thamilarasan et al. [24] used an improved System Usability Scale (SUS) and software quality standards to assess the interface's effectiveness, efficiency, and user satisfaction. Their study used structured Likert-scale questionnaires and showed that standardized usability testing reveals interface bottlenecks such as problematic navigation structures and unclear labels and results in statistically significant improvements in the quality of user interaction following design iterations. The SUS test instrument was particularly well suited for testing systems designed for a broad user population with differing technical skills. Baniaga et al. [25] tested a web-based campus map guide using end-user testers and school stakeholders, confirming the spatial and routing data and descriptions of facilities. Their methodology used structured functional test checklists for students and faculty testers and demonstrated the value of end-user involvement for testing the functional accuracy of spatial navigation systems. For security testing, Manik et al. [13] noted that it's important to test role-based access control (RBAC) boundaries in multi-user academic portals. Their testing

involved manual penetration test cases for authentication and role permission controls and showed that properly enforced RBAC security controls prevented unauthorised access to administrative and sensitive data.

The literature shows that multi-dimensional software testing is important to ensure the quality, security, and usability of web-based systems for campus use. Black-box functional testing of ISO/IEC 25010:2011 criteria and test cases for system features and response to inputs is a good way to test system functions. Validated usability measures like the SUS enable quantitative, objective measurement of user interaction quality and manual testing of RBAC policy enforcement ensures security constraints are correctly applied. These methodologies provide a well-validated framework for testing the system aligned with the functional and non-functional requirements of role-based, PWA-based, campus information systems. These approaches are essential to ensure the proposed system for CSPC is functional, usable and secure before it is deployed.

2.2 Synthesis of the State-of-the-Art

The literature reviewed in these thematic domains shows a clear trend toward improvement in digital systems that relate to the campus, with each domain contributing to one or more of the issues of navigation, access to information, connectivity and user experience. Campus navigation systems that are web-based and non GPS-based, as reviewed by Madhavan et al. [1], Anusiuba et al. [2], and Bhusekar et al. [5], have been reported as successfully decreasing orientation confusion, especially in the case of new students and other high-traffic academic events. These tools are primarily based on 2D map views and searchable directories. The system works as intended but it lacks the ability to deliver actual spatial direction and to create different types of content which match specific user responsibilities. The virtual tour systems which Chiron and Hamidan [6] and Pujari et al. [7] and Amali et al. [8] studied which include 360-degree panoramic navigation enable users to better understand their surroundings and become familiar with campus layouts. The systems provide basic functions which serve promotional and

introductory purposes but they lack essential operational features which users need to navigate their daily activities through start-to-destination routing and role-specific dashboards, dynamic room schedules and campus information management systems.

The literature on Progressive Web Applications (PWA), as reviewed in the contributions of Cherukuri [9], Rao and Rekha [10] and Nugraha et al. [11], show that Progressive Web Application (PWA) designs offer improved performance, offline capability and cross-platform portability over traditional web applications. Such characteristics are vital for the Philippines' educational environments, where Umar and Argungu [16] found that many students have limited and intermittent internet access, and offline systems have been shown to significantly enhance access and educational outcomes. RBAC implementations reviewed by Manik et al. [13], Lavhe et al. [14], and Romli et al. [15] are commonplace in academic portals as means of organising access privileges, minimising information clutter and protecting data; yet most current campus navigation and virtual tour systems are presented without role-based personalisation to all user types. Cloud and academic cloud deployments, reviewed through Sirajuddin [18], Oleksiuk and Oleksiuk [19], and Ouda and Yas [20], provide scalable, centralised cloud resources for managing and sharing campus data, and serving a large number of users. Smart campus frameworks proposed by Polin et al. [21] and Raguro et al. [22] also establish navigation services and information access tools, as well as user-friendly digital systems, as fundamental elements of institutionally mature, technology-supported learning environments. Software quality assurance literature reviewed from Suryadi and Sulistiyani [23], Thamilarasan et al. [24], Baniaga et al. [25], and Manik et al. [13] demonstrates that black-box testing using ISO/IEC 25010:2011 criteria, System Usability Scale (SUS) questionnaires, end-user test checklists, and RBAC security testing, offers a robust approach to the evaluation of educational websites. The synthesis of these literatures makes it evident that there is a convergence point, where it is necessary for campuses to have navigational and information systems that should transcend mere utility-based purposes

and become immersive, context-sensitive, accessible offline, and in line with the concept of smart campuses.

2.3 Gap Bridged by the Study

The reviewed literature shows that current approaches to campus navigation (web-based campus navigation, virtual tour systems, PWA-based systems, and role-based academic portals) solve some aspects of campus information access, but when put together do not solve the spectrum of wayfinding and information management problems that exist in academic settings. Campus navigation systems offer directory and route-based navigation, but these are based on 2D interfaces without immersive spatial orientation, role-based personalisation, and data integration. Virtual tour systems provide 360 degree panoramic views for initial orientation but are not designed for daily use, lacking essential features such as point-to-destination navigation, dynamic room scheduling, and role-based information. PWA systems have excellent offline and cross-platform features, but have not been combined with immersive navigation and role-based information delivery in a campus portal. RBAC technology is already widely adopted for controlling permission in academic portals but has yet to be applied to the campus navigation and mapping domain. Moreover, no campus navigation system explicitly includes offline caching of navigation paths, panoramic views and room data to support offline use in low connectivity contexts. In sum, existing systems are isolated, catering to distinct features without providing immersive navigation, offline capability and role-based content delivery in an integrated system.

To address these limitations, there is a need for an integrated system that combines non-GPS navigation, immersive 360-degree visualization, offline-capable web technologies, and role-based access control in a single platform. This system can offer pre-planned routes complemented by an interactive three-dimensional depiction of space, thus helping individuals better comprehend the layout of the college campus through more than just the standard two-dimensional map. By integrating role-based dashboards into this design, information and services relevant to each individual, whether student, faculty member, administrator, or guest, can be

provided. In addition, progressive web application attributes along with caching techniques would ensure continued access even when network coverage becomes sparse. Through integration of cloud-based and the smart campus framework, such an implementation will be a successful one.

Notes

- [1] Ajay Madhavan K. L. K., Hariharan K. S., Kowshigan M. V., Prasanna Kumar M. N., Sanjay T. S., Parthasarathy S., and Thangasankaran R. 2026. Campus Guidance System. *International Journal for Research in Applied Science & Engineering Technology (IJRASET)* 14, 1 (Jan. 2026), 1832–1840. <https://doi.org/10.22214/ijraset.2026.77223>.
- [2] Overcomer Ifeanyi Alex Anusiuba, Mary Rose Ngozi Umeh, Evelyn Ogochukwu Ezuruka, and Samuel Olujimi Adejumo. 2025. Design and Implementation of Web-Based Digital Campus Navigation and Orientation System for New Student. *International Research Journal of Innovations in Engineering and Technology (IRJIET)* 9, 2 (Feb. 2025), 41–57.
- [3] T. Sharma, A. Gupta, and R. Singh. 2024. Performance Evaluation of Vector-Based Web Mapping Libraries: A Case Study of MapLibre GL JS in Urban Navigation. *International Journal of Web Information Systems* 20, 2 (Mar. 2024), 112–125. <https://doi.org/10.1108/IJWIS-10-2023-0145>
- [4] Nordin N., Markom M. A., Suhaimi F. A., and Ishak S. 2021. A Web-Based Campus Navigation System with Mobile Augmented Reality Integration. *Journal of Physics: Conference Series* 1997, 012038 (2021), 1–8. <https://doi.org/10.1088/1742-6596/1997/1/012038>.
- [5] L. Fernandez and C. Reyes. 2025. Scalable Backend Architectures for Institutional Web Portals Using Node.js and Express.js. *Journal of Applied Web Engineering* 12, 1 (Feb. 2025), 45–58.
- [6] Achmad Choiron and Rusdi Hamidan. 2025. UX Analysis of the Virtual Tour 360 Application at Universitas Dr. Soetomo Campus. *Journal of Applied Informatics and Computing (JAIC)* 9, 1 (Feb. 2025), 73–78.

-
- [7] Priyanka Pujari, Shashidhar Sharan, Sarika Patil, Vaibhav Kalungada, and Sanket Khot. 2025. Designing an Immersive Virtual Reality Campus Tour. *International Journal of Research Publication and Reviews* 6, 5 (May 2025), 14316–14319. <https://doi.org/10.55248/gengpi.6.0525.1955>.
- [8] K. Patel, S. Deshmukh, and M. Lee. 2024. Optimizing Immersive Virtual Tours: Integrating Pannellum.js and Cloud-Based Media Delivery Networks. *Journal of Interactive Educational Media* 15, 3 (Aug. 2024), 210–222.
- [9] Bangar Raju Cherukuri. 2024. Progressive Web Apps (PWAs): Enhancing User Experience through Modern Web Development. *International Journal of Science and Research (IJSR)* 13, 10 (Oct. 2024), 1549–1560. <https://doi.org/10.21275/MS241022095359>.
- [10] Manu S. Rao and Rekha B. S. 2024. Progressive Web Apps in Cross-Platform Development: A Comparative Analysis and Evaluation. *IJSREM* 8, 6 (June 2024), 1–6. <https://doi.org/10.55041/IJSREM35595>.
- [11] Dian Nugraha, Febria Anjara, and Safira Faizah. 2022. Comparison of Web-Based and PWA in Online Learning. In *FIRST–T1–T2 2021 International Conference. Atlantis Highlights in Engineering* 9, 201–205.
- [12] Nivedita Vibhandik, Pranav Kawale, Mayuresh Sonar, Mrunal Divate, and Aditya Sonar. 2025. Hybrid PWA Navigation System for College Campuses. *International Journal of Engineering Technologies and Management Research* 12, 10 (Oct. 2025), 1–6. <https://doi.org/10.29121/ijetmr.v12.i10.2025.1683>.
- [13] Albert Ramadhan Manik, Dedy Kiswanto, Muhammad Budi Akbar, and Jogi Purba. 2025. Academic Portal with MFA (WhatsApp OTP), Role-Based Access Control, and Logging System for Network Monitoring. *Jurnal Ilmiah Sistem Informasi (JUI SI)* 4, 3 (Nov. 2025), 674–687. <https://doi.org/10.51903/qwz6pt87>.

-
- [14] Ramesh Lavhe, A. N. Kalal, Rasika Lokolkar, Shreyas Kulkarni, Siddhesh Dhande, and Vishal Sharma. 2025. Academic Management Through a Web-Based Student Profile System. *IJRASET* 13, 5 (May 2025), 7347–7354.
- [15] Rusnida Romli, Muhammad Afiz Aznan, Lip Zhong Xian, Nur Anis Alia Bakhoruddin, Fatin Nur Hazwani Mohd Wazir, and Amar Raaj Singh. 2021. AR@UNIMAP: A Development of Interactive Map Using Augmented Reality. *Journal of Physics: Conference Series* 1755, 012052 (2021), 1–7. <https://doi.org/10.1088/1742-6596/1755/1/012052>.
- [16] Ibrahim Umar and Kasim Abubakar Argungu. 2025. Offline Capable Mobile Learning Solutions for Low Connectivity Regions. *International Journal of Sustainable Social Science (IJSSS)* 3, 5 (2025), 337–354. <https://doi.org/10.59890/ijsss.v3i5.107>.
- [17] Torsten Braun. 2025. Dual-Engine Intelligent Caching: A Joint Optimization Framework for 360° Mobile VR Video Edge Caching. In *Proceedings of the International Conference on Mobile Computing and Networking (MobiCom '25)*. ACM, New York, NY, USA, 1–12. <https://doi.org/10.48620/86936>.
- [18] Mohammed Khaja Sirajuddin. 2024. Cloud Technologies Revolutionizing Global Education: A Technical Case Study. *International Journal of Scientific Research in Computer Science, Engineering and Information Technology* 10, 6 (Nov.–Dec. 2024), 2084–2092. <https://doi.org/10.32628/CSEIT2410612400>.
- [19] Soham Rahul Joshi, Rushil Jayendra Ekhande, Ibrahim Irfan Kazi, Tanmay Vinayak Naik, and Shubhangi Sanket Chintawar. 2026. Zaprint – A Remote Printing Management System. *International Journal For Multidisciplinary Research (IJFMR)* 8, 1 (February 2026). <https://doi.org/10.36948/ijfmr.2026.v08i01.68894>
- [20] A. M. Hassan and F. Qureshi. 2024. Transitioning from Private Clouds to Serverless Architectures in Higher Education Information Systems. *IEEE Transactions on Cloud Computing* 12, 4 (Oct. 2024), 1544–1556.

-
- [21] Ken Chalapan Polin, Tan Yigitcanlar, Mark Limb, and Tracy Washington. 2023. The Making of Smart Campus: A Review and Conceptual Framework. *Buildings* 13, 4 (Mar. 2023), 891. <https://doi.org/10.3390/buildings13040891>.
- [22] Ma. Corazon G. Fernando Raguro, Ace C. Lagman, and Ronaldo A. Juanatas. 2022. Technology Management Framework for Smart University System in the Philippines. In *ICIT* 2021. ACM, New York, NY, USA, 372–380. <https://doi.org/10.1145/3512576.3512642>.
- [23] Devaldi Akbar Suryadi and Endang Sulistiyani. 2022. Evaluation of Information Quality Using ISO/IEC 25010:2011 (Case Research: Menu Harianku Application). *International Journal of Innovation in Enterprise System (IJIES)* 6, 2 (July 2022), 119–132. <https://doi.org/10.25124/ijies.v6i01.165>.
- [24] Yarshini Thamilarasan, Raja Rina Raja Ikram, Mashanum Osman, Lizawati Salahuddin, and Wan Yaakob Wan Bujeri. 2023. Enhanced System Usability Scale using the Software Quality Standard Approach. *Engineering, Technology & Applied Science Research* 13, 5 (2023), 11779–11784. <https://doi.org/10.48084/etasr.5971>.
- [25] Devonne Clyde G. Baniaga, Gartly R. Cortez, Iordanis Keith D. Jacinto, Miella Carmel F. Rabago, Neil Bryan V. Real, Rein Arwen D. Santos, Jazhmaigne Lizney Ea C. Taylan, Keziah Faye M. Arellano, Johnny R. Pagaduan, and Jhansen Rey M. Obispo. 2023. Development and Testing of a Digital Map Guide of Ilocos Norte National High School Main Campus. In *DLSU Senior High School Research Congress*. De La Salle University, Manila, Philippines. https://animorepository.dlsu.edu.ph/cgi/viewcontent.cgi?article=1852&context=conf_shsrescon

CHAPTER 3

TECHNICAL BACKGROUND

This chapter presents the technical foundation and methodological procedures that will be employed in the development of CampusSphere, a role-based online portal with VR-supported campus mapping implemented as a Progressive Web Application (PWA) for Camarines Sur Polytechnic Colleges (CSPC). It discusses the research design, data sources, instruments, and analytical methods that will be applied during the requirements gathering and system evaluation phases of the study. The chapter also describes the theoretical and conceptual background that will guide the system design, including the selected software development methodology, design patterns, and algorithms. Furthermore, this chapter outlines the proposed system architecture, software and hardware development stack, tools, and platforms that will support the implementation of the system. It concludes with the testing plan, which describes the procedures and criteria that will be used to assess the functionality, usability, security, and user satisfaction of the proposed CampusSphere platform.

3.1 Research Design and Methods

The study will utilize a mixed-method research design that combines qualitative and quantitative methods to support the system development and evaluation process of the proposed CampusSphere system. This approach is appropriate for system development research as it enables the researchers to explore and understand user requirements and system processes in detail, as well as to offer empirical evidence of system efficacy and user satisfaction [1].

The qualitative approach will be used to understand the current practices of campus navigation, information access, and user needs at CSPC. Interviews will be held with students, academics, administrators and visitors to determine functional and non-functional needs for the system. The quantitative component will be used in the system evaluation stage, where various testing tools and evaluation surveys will be administered to objectively measure the system's functionality, usability, security and user satisfaction [2].

3.1.1 Data Sources

This section describes the primary data sources utilized to support the development and evaluation of the system, detailing the sampling technique, sample size computation, and respondent distribution.

Sampling Technique

This research uses **Stratified Mixed Sampling** and Total Enumeration to assess the CampusSphere portal [3]. This hybrid methodology accommodates the system's Role-based Access Control (RBAC) model, which demands both statistical validity on the part of end users and technical verification by administrators. The population is stratified according to system roles, with the appropriate selection method applied to each strata:

Probability Selection for General Roles (Students & Faculty). To ensure an unbiased sample for the System Usability Scale (SUS) and user satisfaction surveys, proportional random selection is used for the Student and Faculty strata. The sample size for statistical significance was calculated using Slovin's Formula [4]. Using a typical value of 5% (0.05) for the margin of error to ensure reliability, the calculations are as follows:

For the **Student Stratum**: Based on official records indicate the current CSPC student population is 13,444 [5].

$$n = \frac{N}{1 + Ne^2}$$
$$n = \frac{13444}{1 + 13444(0.05)^2}$$
$$n = \frac{13444}{1 + 33.61}$$

Because the real-time demographic division of the 13,444 students is fluid, Proportional Allocation was applied. As established in recent methodological literature by Mishra et al. (2023), proportional stratified sampling ensures that the sample size allocated to each stratum accurately reflects its proportional weight within the historical population [6]. Using historical institutional

enrollment estimates of approximately 43% Freshmen/Transferees and 57% Regular Students, the 389 base sample is proportionately allocated into exactly 167 Freshmen and 222 Regular Students.

For the **Faculty Stratum**: Based on the obtained population of 493 active faculty members:

$$n = \frac{493}{1 + 493(0.05)^2}$$
$$n = \frac{493}{2.2325}$$

This establishes a probability sample of 221 faculty respondents. Individual respondents within both the student and faculty strata will be drawn via simple random sampling against official institutional registries.

Total Enumeration for Specialized Roles (Administrators). Because general users do not have the security privilege to access the admin dashboard, the testing will be carried out by the CSPC ICTU personnel. Since the target population (system administrators) is extremely limited (11 individuals), Total Enumeration (census survey) will be used. Saunders et al. (2023) have stated that when the population is very small, total enumeration will be the most accurate as well as eliminating any possibility of selection bias [7]. The entire 11 administrators will be purposely chosen to run the security test cases and assess the back-end modules.

Convenience Selection for Unregistered Roles (Guests). Campus guests and visitors do not form a population frame, therefore probability sampling cannot be employed. As such, an additional convenience sample of 40 real campus visitors will be taken. The sample size was determined using the Central Limit Theorem (CLT). As confirmed in contemporary 32 engineering models by Walpole et al. (2021), $n \geq 30$ is a statistically valid sample size for the sampling distribution to assume a normal distribution, so this is the accepted minimum sample size for parametric analysis of an unknown population [8].

Respondents

This section identifies the individuals who participated in the study. Table 1 presents the distribution of the 661 total respondents included in the study based on their respective roles.

Table 1: Distribution of Respondents

RESPONDENTS	NO. OF RESPONDENTS	PERCENTAGE	JUSTIFICATION
Students (Freshmen & Transferees)	167	25.26%	Prioritized as main respondents because they are the least familiar with the campus environment and will benefit most from the navigation features. (Determined using Proportional Allocation of 43%).
Regular Students	222	33.59%	Provided broader usability feedback from a user base that regularly interacts with various campus facilities. (Calculated via Proportional Allocation of 57%).
Faculty Members	221	33.43%	Contributed expert validation of location data accuracy and scheduling information displayed within the system. (Calculated via Slovin's Formula)
System Administrators (ICTU Staff)	11	1.66%	Assessed the portal's administrative interface, role-management features, and data security controls. (Utilized Total Enumeration).
Guests / Campus Visitors	40	6.05%	To ensure that the guest access features and the 360-degree VR tour are tested by non-members. (Sample size in accordance with the Central Limit Theorem).
TOTAL	661	100%	

Table 1 shows the respondents comprise of five different categories of people who collectively represent the majority of CampusSphere users. Undergraduate first and transferee students were identified as key respondents as they are the target users who are least familiar with the campus and will benefit most from the campus navigation features of the portal. Non-freshmen students offered general usability feedback from a user group that uses a range of campus facilities.

Academics provided expert feedback regarding the accuracy of the location information and the scheduling of events in the system. ICTU staff members with system administrator roles evaluated the portal's administrative interface, role-based access control, and security measures.

Importantly, the category of Guests and Campus Visitors was included in the list of respondents in acknowledgement of CampuSphere's public Progressive Web Application that does not require a user to login. Campus guests, such as parents, prospective students and community partners, often need to find their way around campus and gather information about facilities. Convenience sampling for this category allows the study to validate the guest access features, 360-degree VR campus tour, and public map functions with a group of non-members, enhancing the system's accessibility and usefulness.

Secondary Sources

In addition to primary data gathered from survey participants, the research also leverages secondary data sources to complement and verify the results. This included existing administrative records, maps, building directories and official ICTU publications that provide physical evidence of the campus, buildings and organisational structure. Secondary data can be used to identify patterns, confirm user data, and gain insights into processes [9].

In this instance, the researchers examined existing paper-based campus directories, room-assignment schedules, organizational charts and historical campus maps. These data were used to establish the portal's information architecture, to verify that the mapped locations were complete, and to ensure the role-based access levels reflected the institutional hierarchy. The 34 documentary sources helped improve the accuracy and credibility of the system design by relying on institutional data [10].

3.1.2 Research Instruments

This section outlines the research instruments used to collect qualitative and quantitative data necessary for system development and evaluation. These instruments were selected to

effectively capture user requirements, assess existing navigation challenges, and measure the functionality and usability of CampusSphere.

Interview Guide. Semi-structured interviews will be used to collect detailed information on the difficulties students, faculty and staff face when navigating the campus, and to understand what information is required by campus visitors and guests. To address Specific Objective 1, the guide contains key questions about problems navigating the campus, current limitations in distributing room assignment or schedule changes, and what digital features (such as 360-degree views or offline access) would assist users the most. This tool assists with identifying process and user needs for system development.

System Testing Checklist. The technical assessment will be based on a checklist using black-box testing principles. This instrument has two components: functional testing checklist based on the Functional Suitability characteristic of the ISO/IEC 25010:2011 standard to assess feature correctness and a security testing checklist for manual penetration testing of the Role-Based Access Controls (RBAC) to ensure that access is properly controlled. This employs a binary scale (0=Fail, 1=Pass) to evaluate the system's technical readiness for deployment to the end-users.

Evaluation Survey Questionnaire. A structured questionnaire will be used for end-user testing following system deployment to measure user acceptance and satisfaction. This tool uses the standardised System Usability Score (SUS) to assess interface effectiveness, and a 5-point Likert-scale questionnaire to assess the level of overall User Acceptance Testing (UAT) acceptance and satisfaction across all user roles.

3.1.3 Data Gathering Procedure

The data gathering procedure will be conducted sequentially, strictly utilizing the identified research instruments to achieve the project's objectives.

Requirements Gathering. For the requirements gathering phase, which aligns with the project's goal of determining spatial data and system requirements, the researchers will interview users in person and online. The researchers will conduct interviews using the Interview

Guide with some of the target population. This qualitative approach will reveal spatial data requirements, existing bottlenecks in navigating the campus, and announcement processes needed to correctly organise the system's database structure and user interface, to be used in the Thematic Analysis [11].

Technical System Evaluation. The system testing phase will solely be the responsibility of the researchers and select ICTU staff as system administrators. This group will run a series of functional and security test cases in an isolated environment. The data collection will be conducted using the System Testing Checklist (STC) with only the strict binary choice of 0=Fail and 1=Pass, based on ISO/IEC 25010 and common security practices. This will involve the testers to deliberately invoke certain system operations (e.g., login, role change, and VR scene loading) and to determine if the system will properly return the technical outcome.

Usability and Satisfaction Assessment. Once the technical stability of the system is verified, the assessment of usability and satisfaction will be conducted with the 661 identified respondents, including the stratified samples of freshmen, regular students, faculty members, and guests. They will be provided with an opportunity to explore the deployed portal and complete navigation tasks. For consistency, these end-users will not be given binary test cases or checklists. Rather, the data collection process will be based on the Evaluation Survey Questionnaire (ESQ). After their system interaction, the respondents will provide feedback by answering the 10-item System Usability Scale (SUS) to assess the effectiveness of the system, and a 5-point Likert Scale for User Acceptance Testing (UAT) to determine the overall system acceptability prior to final institutional deployment. To ensure methodological consistency, a strict 'one scale per testing dimension' rule was enforced throughout the data gathering process.

3.1.4 Data Analysis Procedure

The data collected in this study will be analyzed using both qualitative and quantitative methods to systematically support the specific objectives of the proposed system. Data analysis will be carried out using the following statistical and analytical tools:

Thematic Analysis. To fulfill Objective 1, the process will begin with the administration of the Interview Guide to selected respondents (students, faculty, administrators, and visitors). This will identify the specific spatial data, current navigation challenges, and institutional information needed to structure the CampuSphere modules.

Binary Evaluation Scale. The binary scale will be used to quantify the data collected from the functional and security test cases. If a test case is successfully executed by the system, the test case is scored as 1 (Pass) or 0 (Fail) if the system returns an error. A percentage of the total pass rate is used to assess technical readiness.

Table 2: Scale for Functionality Evaluation

RANGE	LEVEL OF FUNCTIONALITY	DESCRIPTION
0.67 – 1.00	Completely Functional	There aren't many flaws or no errors are found in the system; it's ready for deployment. All modules are operational, and both incorrect and correct inputs are distinguished. The test scenarios get the desired result.
0.34 – 0.66	Somewhat Functional	The system is operational, however there are few errors and bugs that require more testing. Can be used based on the end user's preference.
0.00 – 0.33	Least Functional	Not ready for deployment; several flaws and errors have been discovered; additional tests are required. The system is operational, but its functions are limited.

As shown in Table 2, the proposed system's functionality is measured using a scale that defines the system's functionality levels based on the average binary results of the functionality test. This scale allows the researchers to decide whether the system is least functional, partially functional or fully functional based on the result of the computation.

Security Evaluation Scale. The data collected from the manual penetration testing and Role-Based Access Control (RBAC) will also use the binary pass/fail scale (0 = Fail, 1 = Pass). The average pass rate will be used to calculate the system's security status.

Table 3 presents the scale used to evaluate the security readiness of the proposed system.

Table 3: Scale for Security Evaluation

RANGE	LEVEL OF SECURITY	DESCRIPTION
0.67 – 1.00	Highly Secure	The system effectively enforces RBAC constraints and restricts unauthorized access. Critical security boundaries are maintained, making it safe for institutional deployment.
0.34 – 0.66	Partially Secure	The system possesses basic security measures, but vulnerabilities exist within the role boundaries that require immediate patching before full end-user deployment.
0.00 – 0.33	Not Secure	The system fails to enforce access controls, leaving administrative and instructional data exposed to unauthorized users. Extensive system revisions are strictly required.

Table 3 highlights the system's security level based on the successful implementation of the penetration test cases. To achieve a high level of security in the CampusSphere portal, given the nature of institutional data and role management, it is ideal for the portal to be rated under the "Highly Secure" category ensuring the security and quality of the PWA and Supabase database prior to release.

System Usability Scale (SUS) Scoring. For SO3 (system usability), the system's usability will be measured using the standardised System Usability Scale (SUS) scoring system. The SUS questionnaire responses from the end users will be transformed into a composite score ranging from 0-100 to determine the overall baseline of the system's usability and user-friendliness.

The SUS questionnaire's final score will be interpreted according to a standardized scale. Table 4 shows the scale used to assess the acceptability of the system's usability based on the composite SUS score.

Table 4: Scale for Usability Evaluation

RANGE	ACCEPTABILITY LEVEL	ADJECTIVE RATING
85.00 – 100.00	Acceptable	Excellent
71.00 – 84.99	Acceptable	Good
51.00 – 70.99	Marginal	OK
36.00 – 50.99	Not Acceptable	Poor
0.00 – 35.99	Not Acceptable	Worst Imaginable

As shown in Table 4, the computed SUS scores grouped into acceptability levels and adjectives to determine whether the portal's interface, 360-degree VR walkthrough and role-based dashboards are user-friendly for the campus community.

Weighted Mean. For the User Acceptance Testing (UAT) under SO3, the weighted mean will be used to calculate the mean of user satisfaction based on the 5-point Likert scale. This method considers the relative importance of each response category to determine the overall system acceptability. The weighted mean is calculated using the formula.

$$\bar{x} = \Sigma(w_i x_i) / \Sigma w_i$$

Where:

$\bar{x}$ = The computed Weighted Mean

Σ = The Summation operator

w_i = The specific weight assigned to the response category

x_i = The numerical value of the Likert scale response (ranging from 1 to 5)

Table 5 presents the scale used to evaluate the user satisfaction of the proposed system. To ensure statistical validity, the interval ranges were computed using the standard parametric interval formula for a 5-point Likert scale, calculated as $\frac{5-1}{5} = 0.80$, based on the modern methodological framework validated by Kusmaryono et al. (2022) [12].

Table 5: Scale for User Satisfaction Evaluation

Range	Scale	Description
4.21 – 5.00	Strongly Agree / Excellent	The system is highly satisfactory and exceeds expectations.
3.41 – 4.20	Agree / Good	The system is satisfactory and meets user expectations effectively.
2.61 – 3.40	Neutral / Average	The system is moderately satisfactory but has noticeable areas for improvement.
1.81 – 2.60	Disagree / Poor	The system is unsatisfactory and limits the user's ability to achieve goals.
1.00 – 1.80	Strongly Disagree / Very Poor	The system is highly unsatisfactory and fails to meet user requirements.

As presented in Table 5, the user satisfaction of the proposed system is measured using a 5-point Likert scale that reflects the level of acceptance and overall effectiveness of the system. The scale categorizes user satisfaction based on the calculated weighted mean from user responses.

3.2 Theoretical or Conceptual Background

This part discusses the theoretical and conceptual foundations that guide the design and development of CampusSphere. It explains the technologies, frameworks and theories brought together in the portal to ensure speed, accessibility, security and user-friendly navigation.

CampusSphere employs four theoretical concepts: Progressive Web Application (PWA) technology, Role-Based Access Control (RBAC), 360-degree Virtual Reality (VR) navigation and the Smart Campus concept. These concepts constitute the theoretical foundation that guided the system architecture, features and user interface design. The fundamental design principle which governs all four concepts states that CampusSphere allows all users to access its public features without requiring them to create accounts or download software, which results in complete accessibility for every person on campus.

Progressive Web Application (PWA). A Progressive Web Application (PWA) represents a software development approach which developers use to build applications that function as native mobile apps through web browsers by utilizing contemporary web development methods. The system operates between a website and a mobile application by utilizing Service Workers to manage its background tasks and create offline storage capabilities. The theoretical framework of the system enables users to access the campus map through any standards-compliant browser because they do not need to download the application from an app store or install it on their device. The technology enables users to access the map without needing internet access and complete their navigation because it does not require specialized equipment or advanced computing devices. [13]

Role-Based Access Control (RBAC). The security model establishes access control rules which determine system entry rights according to user roles within the organization. The theoretical

model restricts user access rights by connecting access permissions to particular roles which include Admin, Student, Instructor and Guest roles. The system establishes secure information distribution through role-based access control which keeps sensitive content and administrative tasks secure from unauthorized users while allowing public access to basic navigation tools. The system employs role-mapping to separate users from direct access rights which enables it to handle complex team environments while safeguarding institutional data from unauthorized access. This concept is fundamental to maintaining the integrity and scalability of the institutional data within the portal. [14].

360-Degree VR Navigation. The theory of 360-Degree VR Navigation is based on the concept of Immersive Telepresence, where users can perceive they are at a distant place by visual stimuli. The system uses equirectangular images and interactive hotspots to create a virtual environment that resembles the actual campus. This theoretical navigation method expands upon traditional 2D maps, providing a sense of spatial awareness of the buildings and facilities of CSPC, which is important for first-time visitors and students to get acquainted with the environment. By using the spherical panorama images as the basis of the virtual reality environment, the system provides the user with digital mobility to virtually navigate the campus at any time and place. This immersive learning experience is an effective way of informing users of campus facilities and services, while offering an enjoyable and environmentally- friendly method of wayfinding. [15].

Smart Campus Concept. The Smart Campus paradigm is the conceptual framework underlying this research. It refers to the integration of Information and Communication Technology (ICT) and physical infrastructure of an educational environment to increase the effectiveness of the campus services and enhance user experience. Theoretically, a smart campus is centred on "ubiquitous access" to information about facilities, announcements and navigation through digital media. This paradigm addresses the complexity of navigating large educational environments by providing a gap between the outdoor map and indoor route, such as hallways and floor plans. CampusSphere is consistent with this paradigm by digitalising the navigation and by aggregating

information about the campus in a single, accessible portal to avoid confusion and time-wasting for the community [16].

This paradigm addresses the complexity of navigating large educational environments by providing a gap between the outdoor map and indoor route, such as hallways and floor plans. CampusSphere is consistent with this paradigm by digitalising the navigation and by aggregating information about the campus in a single, accessible portal to avoid confusion and time- wasting for the community. [17].

3.2.1 Software Development Methodology

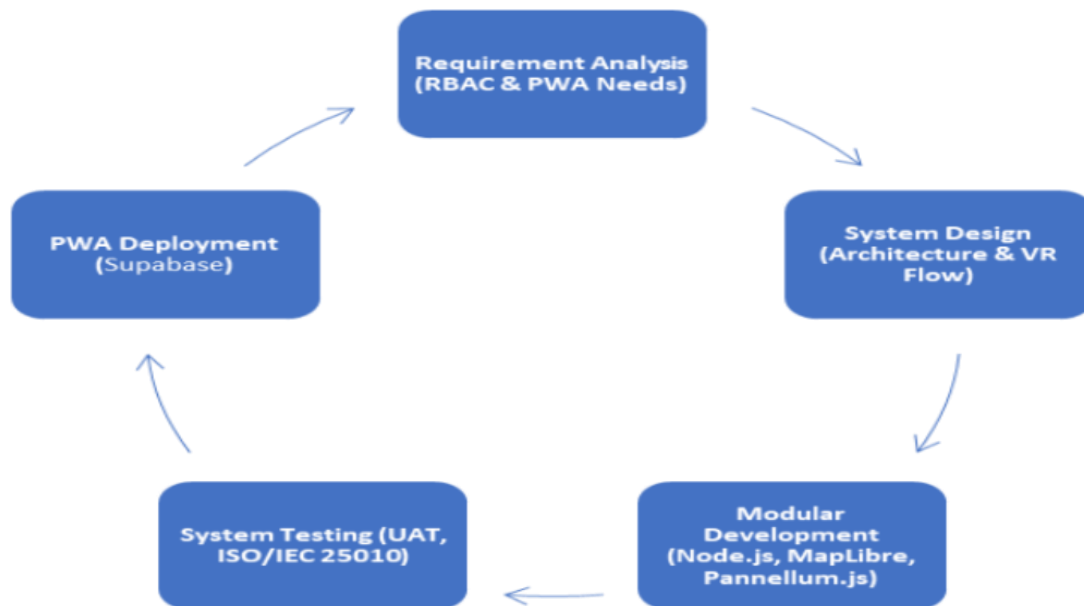


Figure 1. Agile Development Methodology

The researchers will adopt the Agile Development Methodology for the project management, and implement Objective 2, the design and development of the CampusSphere role-based Progressive Web Application (PWA) modules. Agile was selected to enable iterative development and refinement with feedback from stakeholders. This is particularly important for incorporating immersive technologies such as 360-degree VR campus navigation, where prototyping and experimentation will be required to enhance user experience in the virtual space.

Through Agile, the proponents will be able to manage the technicalities and also evolve the spatial design of the system through real-world assessments. [11].

The development will follow these specific phases, tailored to the implementation of the CampusSphere:

Requirement Analysis: Identifying the different requirements and data needs for the campus navigation and announcement module, specifically mapping the Role-Based Access Control (RBAC) roles for Admins, Students, Instructors and Guests using the institutional requirements of CSPC.

Design: The system design requires the development of Entity Relationship Diagrams (ERD) for room scheduling, user management and UI wireframes which will focus on creating a mobile-first Progressive Web Application (PWA). The current stage requires the determination of interactive components and user path analysis through 360-degree VR images which utilize official campus maps.

Development (Sprints): The coding phase is divided into modular sprints where we are applying the identified technology stack to build the core system modules:

- ***Sprint 1: (Backend & Security):*** The Node.js and Express.js backend system setup includes database connection through Supabase which operates on PostgreSQL database system. The database system uses normalized Entity Relationship tables design to manage user account roles and room scheduling information. The implementation of Google OAuth RBAC middleware enables secure management of user accounts.
- ***Sprint 2: (Mapping & VR):*** The development process includes creating an interactive campus map through MapLibre GL JS, which will work together with the 360-degree panoramic virtual reality navigation system built on Pannellum.js. The system uses Cloudinary as a secure platform for storing and optimizing high-resolution equirectangular images

- ***Sprint 3: (Information & Offline Access):*** The project involves three main tasks which include building role-based dashboards, developing the announcements module and implementing PWA Service Workers to provide offline access to essential map assets and routes.

Testing: To achieve Objective 3, the team will perform unit, integration, and User Acceptance Testing (UAT) using the ISO/IEC 25010:2011 standards and the System Usability Scale (SUS) to ensure that the VR navigation, offline mode, and dashboards for Admins, Students, Instructors, and Guests are secure and work effectively for all user roles. **Deployment & Maintenance:** Database hosting on Supabase, 360-degree VR images are stored via an asset delivery network (e.g. Cloudinary) to reduce bandwidth usage, and the app is hosted via a cloud platform (e.g. Vercel) to ensure availability and the mobile experience, meeting Objective 4.

Deployment & Maintenance: The database operates through Supabase while 360-degree VR images use an optimized asset delivery network that Cloudinary provides for their storage needs and the application runs on a contemporary cloud solution through Vercel which delivers high uptime and mobile accessibility, thus completing Objective 4.

3.2.2 Design Patterns, Algorithms, and Core Technologies

The section examines the design patterns and algorithms which CampuSphere uses to achieve secure and efficient operations while maintaining user-friendly system performance.

Model-View-Controller (MVC) Architectural Pattern. CampuSphere implements a keyword-based search algorithm to help users quickly locate campus facilities through partial or full-text queries. The algorithm retrieves relevant information by scanning the database and organizing results based on relevance and specific ranking factors [18]. The system uses exact text matching together with fuzzy text matching to enable users to find records with maximum accuracy through different input methods.

Role-Based Access Control (RBAC) Algorithm. To manage permissions of the users, Role Based Access Control (RBAC) will be used as an algorithm that categorizes permissions under

roles so that the principle of least privilege can be maintained. The portal requires RBAC implementation because it needs to manage user access during real-time events and user interactions that occur throughout the system [19]. Through RBAC the proponents restrict access to sensitive room allocation information and logs to authorized personnel who hold Admin and Instructor roles.

Keyword-Based Search Algorithm. CampuSphere uses a keyword search system which enables users to find campus facilities through their search of complete or partial text. The algorithm retrieves relevant information by scanning the database and organizing results based on relevance and specific ranking factors [20]. The system achieves complete record visibility through its support of both exact text matching and fuzzy text matching which enables users to input different types of search queries.

Pathfinding Algorithm (Dijkstra's or A* Search). The implementation of CampuSphere's interactive mapping module requires a pathfinding algorithm which will use either Dijkstra's algorithm or A* Search. The two algorithms function as fundamental tools which help users navigate between different parts of complex environments. The algorithm calculates the best path which starts from the user's current position and leads to their chosen destination by identifying the least resource-intensive route. [21] Such an approach is imperative for real-time navigation with maximum reliability and accuracy.

Service Workers (Caching Algorithm). Since CampuSphere is a Progressive Web App, it will utilize Service Workers to manage caching and asynchronous data processing. The main component of PWAs enables users to access applications without internet connection while maintaining application performance standards that the project aims to achieve according to its innovation goals (SDG 9) [13]. The system works smoothly because it follows best practices which allow operations to continue without needing permanent internet access, thus creating a system that handles data more efficiently.

PostGIS Spatial Extension. The platform utilizes PostGIS capability to manage spatial data which is essential for creating an interactive map and virtual reality hotspots. The PostGIS extension enables PostgreSQL databases to store and oversee latitude-longitude information which is used to create MapLibre GL JS building nodes and Pannellum.js scenes in the Supabase backend database.

3.2.3 Cryptographic Hashing Algorithms

The system uses Google OAuth 2.0 for authentication and session protection which employs cryptographic hashing algorithms like SHA-256 to protect user session tokens and validate credentials [22]. The algorithms enable access control protection through their ability to create permanent secure hashes from sensitive authentication information which protects Role-Based Access Control (RBAC) boundaries from unauthorized access [23]. The system implements input validation together with data-sanitization processes to defend its administrative backend from common security threats that include SQL injection and cross-site scripting (XSS) attacks [24].

3.3 System Architecture

CampuSphere system operates through a Client-Server Architecture which includes three components: user interface, application logic, and database components. The system design allows users to access the system through web-based clients which include the Progressive Web App while the system maintains its essential functions on the main server. The CampuSphere system achieves scalability and maintainability while retaining its system reliability through its functional distribution across different system layers.

The client layer serves as the main interface through which users access the system. The system supports four distinct user groups which include students, faculty members, system administrators, and campus guests. The system performs multiple functions which include handling forms, displaying maps, loading 360-degree virtual reality environments, and showing dashboards according to the user's access level. The clients establish secure connections through encrypted HTTPS channels to send their requests to the server. The application server handles all core

business processes which include RBAC implementation, location data retrieval, and user session management from its central position. The database server operates with Supabase (PostgreSQL) as its base system and serves as the complete storage solution for the entire organization. The system uses relational databases to create defined RBAC limits while maintaining user data and campus location information and room schedule data. The system stores all major VR assets which include panoramic 360-degree images on a media content delivery network.

Figure 2 illustrates the system process flowchart of CampuSphere, showing the step-by-step interaction of users with the portal from initial access through navigation and exit.

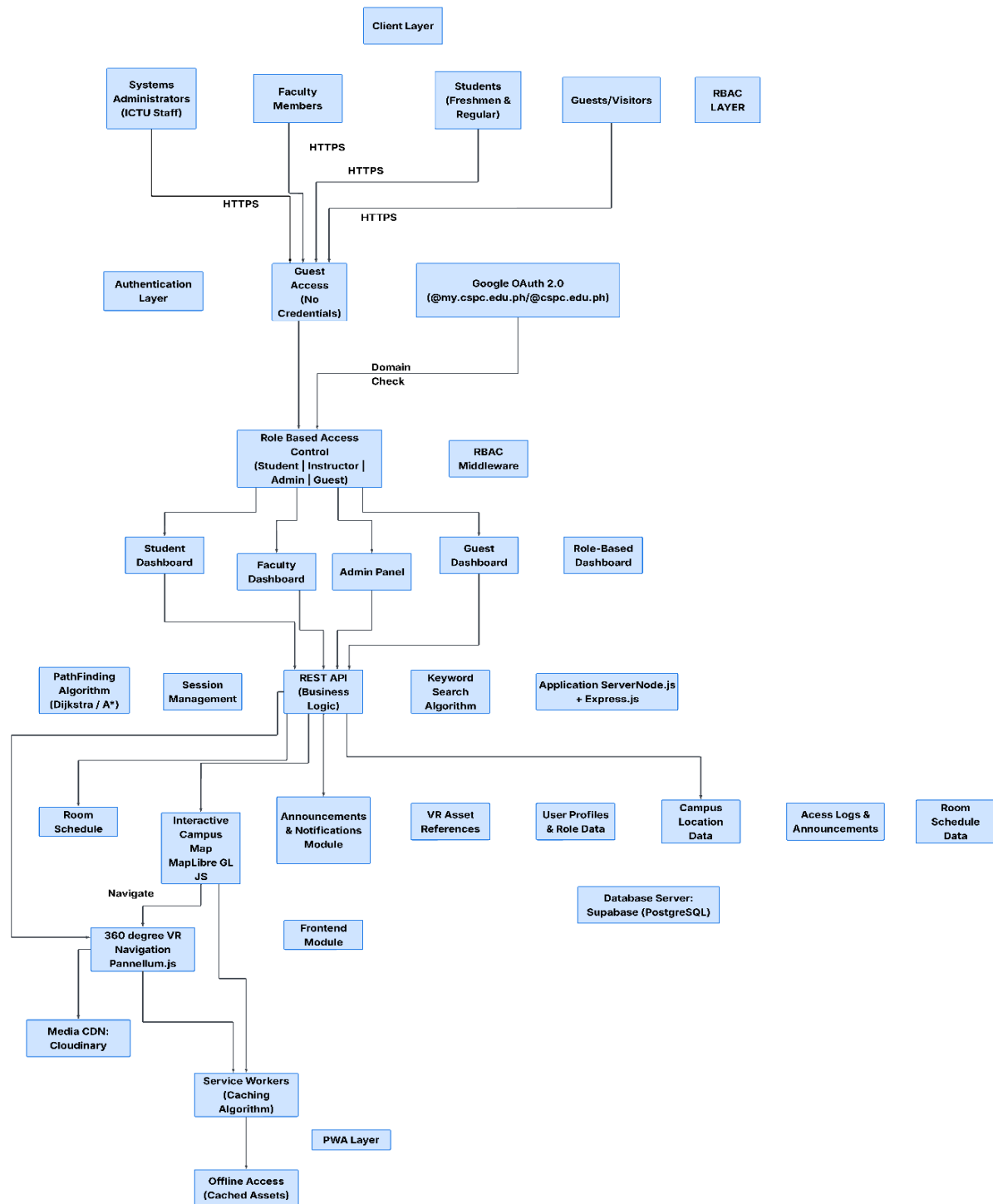


Figure 2. CampuSphere System Architecture Diagram

As illustrated in Figure 2, The CampuSphere system architecture consists of multiple functional layers which interact with each other through secure HTTPS connections. The interaction starts at the Client Layer because different user groups (Administrators, Faculty, Students, and Guests) use the portal to access the system. The Authentication Layer processes incoming requests which use Google OAuth 2.0 to authenticate institutional domains or provide Guest Access without credentials. After authentication, the RBAC Middleware assesses the user's access rights to direct them towards their designated Role-Based Dashboard.

From the dashboards, the user actions initiate REST APIs which establish communication with the Node.js and Express.js based Application Server. The server operates essential business functions through its implementation of keyword search systems and schedule organization and pathfinding algorithm capabilities. The server provides real-time data updates to Frontend Modules which display the interactive campus map through MapLibre GL JS and create 360-degree virtual reality navigation scenes using Pannellum.js. The Supabase PostgreSQL Database Server safely handles all permanent system information which includes user profiles and campus locations and standardized room schedules. The dedicated external asset network delivers high-resolution VR asset images while the system stores their URL references and spatial coordinates in relational databases. The system architecture functions through a PWA Layer which uses Service Workers to store essential rendering components while enabling portal usage throughout offline periods that occur during network interruptions.

3.4 Software Development Stack

This section presents the software technologies utilized in the development and implementation of CampuSphere. The selected software components deliver the necessary technical infrastructure which enables the development of a role-based online portal system that includes Progressive Web Application (PWA) features and complete campus navigation.

Table 6 presents the software development stack specification used in building the proposed system.

Table 6: Software Development Stack Specification

COMPONENT	TECHNOLOGY	VERSION
Operating System	Windows 10 / 11	Build 22H2 / 23H2
JavaScript Runtime Environment	Node.js	v20.11.x LTS
Back-end Framework	Express.js	v4.18.x
Front-end Technology	HTML5, CSS3, JavaScript	W3C Living Standard
Interactive Campus Map Library	MapLibre GL JS	v4.1.x
360-Degree Panoramic VR Library	Pannellum.js	v2.5.x
Database & Authentication	Supabase (PostgreSQL)	PostGIS Extension enabled
Asset & Media Hosting	Cloudinary (or equivalent CDN)	Free Tier API

As shown in Table 6, CampuSphere uses its current JavaScript technology stack together with its PWA capabilities to create systems that grant users access based on their specific roles. The system's back-end operates on Node.js which functions as a non-blocking event-driven runtime that processes multiple simultaneous requests. The back-end framework of the system uses Express.js to create organized API paths that handle user authentication and deliver data according to user roles. The front-end uses standard web technologies HTML5 CSS3 and JavaScript to create a design that works on most web browsers. The system uses MapLibre GL JS to create interactive campus maps through vector-based rendering and Pannellum.js to display 360-degree panoramic scenes. The system uses a Supabase PostgreSQL relational database to securely store user profiles and room schedules while implementing data access control through role-based access control. Cloudinary serves as the dedicated media delivery network which hosts large 360-degree VR assets to optimize bandwidth usage. Service Workers enable the system to store critical assets in cache which allows users to use the system at limited capacity during times of unstable network access.

3.5 Hardware Development Stack

This section details the basic hardware components that are necessary for end- users (students, faculty, administrators, and guests) to be able to use and engage with the CampuSphere

portal effectively. Since the system is implemented as a cloud-hosted Progressive Web Application (PWA), it does not need the on-premise institutional servers. On the other hand, the hardware specifications here are mainly about the client-side processing power, memory, and storage required for a smooth rendering of WebGL-based interactive maps and 360-degree virtual reality scenes within a web browser. Table 7 presents the minimum hardware specifications required to run the system.

Table 7: Hardware Development Stack Specification

COMPONENT	TECHNOLOGY	VERSION
Desktop / Laptop Client	Intel Core i5 Processor (or AMD equivalent)	10th Generation
Desktop / Laptop Memory	DDR4 RAM	8 GB
Local Storage (For PWA Caching)	Internal Device Storage (HDD/SSD)	500 MB Free Space
Network Connection	Wi-Fi or 4G LTE Mobile Data	5 Mbps Minimum
Mobile Device	Android / iOS Smartphone	Android 11 / iOS 12 / 4 GB RAM

As shown in Table 7, the hardware requirements are designed to be lightweight and accessible to accommodate the diverse devices owned by the CSPC campus community. For desktop and laptop users, a basic multi-core processor (such as an 10th Generation Intel Core i5) with at least 8 GB of RAM is required to prevent browser memory exhaustion when rendering the MapLibre GL JS vector graphics and the Pannellum.js 360-degree panoramas. For mobile users, an Android or iOS smartphone with a minimum of 4 GB of RAM is necessary, as high-resolution panoramic images demand significant client-side graphics processing. Furthermore, while the PWA is capable of offline access, a baseline network speed of 5 Mbps is recommended for the initial caching phase, alongside at least 500 MB of free local storage on the user's device to securely store the service worker caches and offline map data.

3.6 Tools and Platforms

This section details the development tools and platforms utilized throughout the design, implementation, and testing phases of CampuSphere. These tools facilitate efficient coding, version

control, and secure access management, ensuring adherence to established software engineering practices.

Table 8 presents the development package specifications used in the implementation of the proposed system.

Table 8: Development Package Specification

PACKAGE / LIBRARY	VERSION	PURPOSE
Visual Studio Code	v1.89.x	IDE used to develop the system
GitHub	v2.44.x	Version control system for tracking source code changes
Google OAuth	2.0	API used for secure @cspc.edu.ph and @my.cspc.edu.ph authentication

Based on the information presented in Table 8, development of CampusSphere with a unified set of platforms support the lifecycle of the Agile software development. Visual Studio Code is chosen as the Integrated Development Environment (IDE) because of its vast extension ecosystem and natural compatibility with JavaScript workflows. GitHub is used for version control, which allows the research team to manage development sprints in iterations, track the changes in code, and keep a reliable history of the repository. Besides, system security and role assignment are done through the Google OAuth 2.0 API. This API only authenticates official portal access on campus using CSPC institutional accounts, sending faculty and administrators through the @cspc.edu.ph domain, and students who are enrolled through the @my.cspc.edu.ph domain. Users who do not have these credentials are automatically redirected to the Guest Role dashboard.

3.7 Testing Plan

This section outlines the testing procedures designed to address Specific Objective 3, ensuring CampusSphere functions accurately and meets specified user requirements. The system evaluation phase is strictly structured around four dimensions. Technical validation will be conducted using the System Testing Checklist (STC) for Functional and Security Testing. End-user perception will be evaluated using the Evaluation Survey Questionnaire (ESQ), which incorporates the System Usability Scale (SUS) for Usability Testing and a 5-point Likert scale for User

Acceptance Testing (UAT). This approach utilizes only the methodologies validated in the Review of Related Literature.

3.7.1 Development Testing Plan

The development testing plan describes the organized steps carried out during the system evaluation phase to verify that CampuSphere meets all functional, security, and usability requirements as specified before it goes live. The development testing flowchart depicts that this methodical procedure is split into four main stages. The first one, Test Preparation, is where the ISO/IEC 25010:2011 quality characteristics and the expected results are determined [25]. Next, Technical Evaluation is a phase of thorough black-box functional and security testing of the main modules, including PWA, interactive maps, and role-based dashboards. After technical stability is approved, the system goes to End-User Evaluation, where the target users evaluate the system's ease of use and satisfaction with the help of standardized tools such as the System Usability Scale (SUS) and Likert scale surveys. In the end, the Analysis & Deployment phase is concerned with analyzing the collected statistical information, fixing the problems found, and completing the CampuSphere site for institutional use.

Figure 3 illustrates the development testing plan implemented during the system development process.

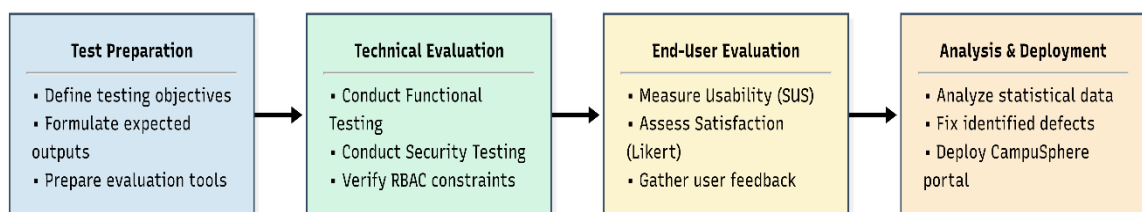


Figure 3. **Development Testing Plan Flowchart**

Based on Figure 3, the development testing plan consists of a series of structured testing stages conducted to ensure that the system functions correctly prior to deployment. The process begins with Test Preparation, where evaluation tools, structured test cases, and expected outputs are established according to ISO/IEC 25010:2011 standards. This is followed by a Technical

Evaluation, which utilizes black-box functional and security testing to verify the performance of core components, such as the PWA modules, interactive maps, and role-based dashboards. An End-User Evaluation is then performed to assess system usability and user satisfaction by gathering feedback through standardized instruments like the System Usability Scale (SUS) and Likert scale questionnaires. Finally, the Analysis & Deployment phase ensures that statistical data is evaluated and any identified defects are resolved, guaranteeing that the CampusSphere portal meets all required functional specifications before its official implementation.

3.7.2 System Evaluation

The system evaluation phase assesses the overall effectiveness and operational readiness of the proposed CampusSphere portal through technical performance and user experience during real-time operations. To make sure that the system very tightly follows the set requirements, this phase is deliberately split into four separate areas, namely: Functional testing (determined by following the ISO/IEC 25010:2011 software quality standard) Usability testing (by means of the validated System Usability Scale) Security testing (done in conjunction with the manual penetration test cases) User Satisfaction testing (through 5-point Likert scale survey) All these formally organized tests put together ensure the integrated modules including PWA, VR navigation, and role-based dashboards work efficiently, are secured and offer a friendly user interface for all kinds of users before institutional rollout.

Functionality Testing. To evaluate the system's functional suitability, black-box testing will be conducted using the ISO/IEC 25010:2011 software quality standards. Structured test cases will be executed to verify that the Progressive Web Application (PWA) modules, interactive map rendering, and role-based dashboards operate exactly as specified. Expected outputs will be compared against actual system behavior to identify and resolve defects.

Table 9: Functionality Testing Test Cases

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
User Authentication Module		
Student Domain Login	Authenticate using a valid @my.cspc.edu.ph account via Google OAuth.	User successfully logs into the designated role-specific dashboard.
Faculty/Admin Login	Authenticate using a valid @cspc.edu.ph account via Google OAuth.	User successfully logs into the designated role-specific dashboard.
Unofficial Email Rejection	Attempt to log in using a non-institutional personal email address	Access to official dashboards is denied; user is redirected to the Guest Role.
Guest Button Access	Select the "Guest" option on the login interface	User is directly routed to the Guest Role dashboard without credentials.
Role-Based Dashboard Module		
Student Dashboard Content	Log in as a student and navigate to the dashboard	Information is strictly limited to student-relevant data.
Instructor Dashboard Content	Log in as a Instructor and navigate to the dashboard	System accurately displays instructor-specific data, such as assigned room schedules and faculty announcements, without granting administrative controls.
Admin Panel Access	Log in as a system administrator and access the administrative backend.	System grants full access to administrative tools, including map content management, system logs, and global campus announcements.
Interactive Campus Map Module		
Building Search	Input a specific campus building name in the map search field.	The system accurately locates and visually highlights the target building on the interactive map.
Map-to-VR Transition	Click the "Navigate" button from a building details panel	The system successfully launches the associated Pannellum.js 360-degree panoramic scene.
Building Hotspot Selection	Interact with an informational hotspot or building marker within the VR scene.	The system displays a details panel containing specific facility and schedule information.
VR Navigation Module		
Panoramic Scene Load	Initiate a navigation route from the main campus gate to the specified building.	The system retrieves the equirectangular asset and successfully renders the panoramic scene, allowing full 360-degree interactive panning without graphical distortion or console errors.
Hotspot-Based Route Progression	Click a directional hotspot within the Pannellum.js scene to advance to the next waypoint	Next panoramic scene in the route loads smoothly.
Route Completion	Follow all hotspot-linked scenes from the starting point to the selected destination building.	User successfully reaches the destination in the VR view.
Announcements and News Module		
News Feed Display	Navigate to the announcements section from any authenticated dashboard.	Announcements are retrieved from the database and displayed chronologically.
Admin Announcement Post	Log in as an administrator and post a new campus announcement.	Announcement is published and visible to all specified users.
Instructor Announcement Post	Log in as an instructor and publish a class-related announcement.	Announcement is published and visible to the relevant student dashboards.
Offline PWA Module		
Offline Cache Availability	Disconnect the test device from the network and access a previously loaded map.	Cached panoramic assets and MapLibre tiles remain accessible.
Offline Mode Indicator	Access the system while offline on the installed PWA application.	System clearly alerts the user that it is operating in offline mode.

Based on Table 9, the functional test cases are structured to execute the black-box testing process outlined in Figure 2. This comprehensively evaluates the performance of all major system

modules from an end-user perspective, including user authentication, role-based dashboard delivery, interactive mapping, VR panoramic navigation, and offline PWA accessibility. Each test case assesses whether the system responds correctly to user actions by verifying successful task execution without inspecting the underlying code. Examples of this include accurately routing users to designated dashboards based on their institutional email domain, correctly rendering sequential 360-degree scenes through Pannellum.js directional hotspots, and ensuring administrative updates are properly reflected portal-wide. By directly comparing actual system behavior against these predefined expected outputs, the researchers can objectively measure functional suitability. Ultimately, the results of these tests will serve as the primary basis for fulfilling the functional testing dimension of Specific Objective 3, ensuring the CampusSphere portal operates in strict accordance with the ISO/IEC 25010:2011 quality standard prior to its deployment.

Usability Testing. Usability will be measured utilizing the standardized System Usability Scale (SUS) instrument. End-users will be tasked with operating the CampusSphere portal, which includes navigating the 360-degree VR controls, plotting campus routes, and accessing role-specific information on their dashboards. Following the completion of these tasks, the SUS questionnaire will be administered to capture quantitative data regarding interface efficiency, navigability, and overall ease of use. This ensures the portal is accessible to the CSPC community regardless of their technical proficiency levels.

Table 10 presents the standard 10-item SUS questionnaire that will be used to assess the user interface. Responses will be measured using a 5-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree).

Table 10: Usability Testing Test Cases

Item No.	SUS Questionnaire Statement	Measurement Scale
1	The system would be used frequently.	5-point Likert Scale
2	The system was unnecessarily complex.	5-point Likert Scale
3	The system was easy to use.	5-point Likert Scale
4	The system requires the support of a technical person to be used effectively.	5-point Likert Scale
5	The various functions in this system were well integrated.	5-point Likert Scale
6	There was too much inconsistency in this system.	5-point Likert Scale
7	The system would be learned very quickly by most people.	5-point Likert Scale
8	The system was very cumbersome to use.	5-point Likert Scale
9	The system inspires confidence during use.	5-point Likert Scale
10	The system requires a lot of prior learning before it can be used efficiently.	5-point Likert Scale

Based on Table 10, the usability test cases are designed to assess how easily users interact with the different modules of CampuSphere. The testing includes VR navigation controls, user interface for authentication, role-based dashboard interface, system behavior in offline mode, and administrator's content management. Specifically, the assessment is focused on the clarity of the

interface, access to the 360-degree VR navigation and the clarity of role-specific information. The tests ensure users can navigate the system effectively and perform essential tasks, such as tracing a campus route, and managing roles without difficulty. The expected outputs will show whether the system supports user-friendly interaction for all roles, leading to better operational efficiency and user satisfaction, as indicated by the System Usability Scale (SUS).

Security Testing. Using manual penetration test cases, will be used to verify the Role-Based Access Control (RBAC) system. This includes attempts to bypass access control, access higher privilege dashboards (e.g., a student trying to access an administrator dashboard), and session token manipulation, ensuring the access control policy is enforced.

Table 11 presents the security test cases that will be used to evaluate the access control and authentication mechanisms of the proposed system.

Table 11: Security Testing Test Cases

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
User Authentication Module		
Unauthorized Admin Access Attempt	While authenticated as a Student, attempt to directly access the administrator dashboard URL.	Access is denied; the system redirects the user to the appropriate student dashboard.
Google OAuth Domain Bypass Attempt	Attempt to authenticate using a non-CSPC email address through the Google OAuth login.	Authentication is rejected; the user is redirected to the Guest Role without portal access.
Role-Based Access Module		
Role Boundary Enforcement-Instructor vs. Admin	While authenticated as an Instructor, attempt to access admin-exclusive content management tools.	Access is denied; instructor-level restrictions are correctly enforced.
Guest Access Boundary	As a Guest user, attempt to access authenticated-only routes (e.g., student dashboard URL).	Access is denied; the system redirects the user to the public guest interface.
Data Integrity Module		
SQL/XSS Injection Sanitization	Attempt to input malicious SQL queries (e.g., 'OR 1=1 --) and basic cross-site scripting (XSS) payloads (e.g., <script>alert('x')</script>) into the login and search input fields.	The system correctly sanitizes the inputs, neutralizes the malicious syntax, and prevents unauthorized database manipulation or script execution.
Unauthorized Announcement Modification	As a Student, attempt to submit a POST request to the announcement creation endpoint.	The server rejects the request with an appropriate authorization error response.
Session Token Validation	Attempt to access a role-protected route using an expired or manually modified session token.	Access is denied; the system invalidates the session and prompts re-authentication.
Vertical Privilege Escalation (RBAC)	Attempt to access a restricted Administrative URL (e.g., /admin/dashboard or /settings/faculty-management) while logged in with a Student or Guest account.	The system's middleware detects the unauthorized role, denies access to the route, and redirects the user to the appropriate low-level dashboard or an "Access Denied" page.

Based on Table 11, the security test cases are designed to systematically verify that the RBAC framework of CampuSphere prevents unauthorized cross-role access and correctly enforces domain-restricted authentication boundaries. The test cases covers key security scenarios such as navigating to pages through direct URL manipulation, accessing API endpoints without proper authentication, attempts to bypass domain restrictions using OAuth authentication, and tampering with session tokens. Each test shows that the access control in the system operate as specified, and that sensitive administrative and instructional data are protected from unauthorised access before the system is deployedat Camarines Sur Polytechnic Colleges.

User Satisfaction Testing. Overall, user satisfaction will be measured through a Likert-scale user satisfaction survey. Following their experience with the CampuSphere, representative of all user groups (Students, Faculty, Administrators, and Guests) will assess their satisfaction with the system's usefulness, clarity of information, ease of VR navigation and offline accessaccessibility.

Table 12 presents the user satisfaction evaluation criteria that will be used to measure the overall user experience of CampuSphere.

Table 12: User Satisfaction Testing Criteria

(1-Strongly Disagree to 5-Strongly Agree)

CRITERION	SURVEY STATEMENT	MEASUREMENT SCALE
System Usefulness	I found the system effective in helping me navigate the CSPC campus and locate facilities.	5-point Likert Scale
Ease of Use	I found the system easy to use without requiring extensive instructions.	5-point Likert Scale
Information Clarity	I found the campus information, room details, and announcements presented in the system to be clear and accurate.	5-point Likert Scale
VR Navigation Satisfaction	I found that the 360-degree VR navigation feature made it easier to understand routes to campus facilities.	5-point Likert Scale
Offline Accessibility	I was satisfied with the system's ability to function under limited or intermittent network connectivity.	5-point Likert Scale
Role-Specific Content	I found the dashboard content displayed to be relevant and appropriate for my specific user role.	5-point Likert Scale
Overall Satisfaction	Overall, I am satisfied with the CampuSphere portal as a campus navigation and information tool.	5-point Likert Scale

Based on Table 12, user satisfaction evaluation criteria are designed to gather information about respondents' general level of satisfaction with CampuSphere in the that are most important to the achievement of the system's main objectives: assisting in navigating the campus, providing information, immersive VR usability and offline functionality. The criteria reflect the major features of the system as identified in the title proposal and Chapter 1 to allow user satisfaction responses to be translated directly as indicators of the intended use. Through the use of a 5-point Likert scale, the data gathered from the user satisfaction survey will be complemented by the results of the functional, usability and security evaluation to give a comprehensive assessment of CampuSphere's effectiveness and readiness for institutional deployment in the Camarines Sur Polytechnic College.

Notes

- [1] Hamed Taherdoost. 2022. What are different research approaches? Comprehensive review of qualitative, quantitative, and mixed method research. *Journal of Management Science & Engineering Research* 5, 1 (2022), 53–63. <https://doi.org/10.30564/jmsr.v5i1.4538>
- [2] Ronaldo R. Canlas, Juan D. Cruz, Maria A. D. Catipon, and Luis R. F. Dizon. 2021. An ISO/IEC 25010 based software quality assessment of a faculty research productivity monitoring and prediction system. In *Proceedings of the 9th International Conference on Information Technology: IoT and Smart City (ICIT 2021)*. ACM, 1–5. <https://doi.org/10.1145/3501409.3501571>.
- [3] Wahyu K. Mahendra and Projo Danoedoro. 2024. Understanding the Influence of Different Sample Sizes and Sample Techniques on Accuracy Assessment of Land Cover Mapping: Case Study of Salatiga city, Indonesia. *Proc. of SPIE*. 12977, 129770E (2024), 1-10. <https://doi.org/10.1117/12.3009445>
- [4] Nurkholis Nurkholis, et al. 2024. Digital Survey for Customer Satisfaction of Regional Drinking Water Companies (PDAM) using the mWater Application and the Slovin Formula Method. *BIO Web of Conferences* 144, 03006.
- [5] Camarines Sur Polytechnic Colleges. 2026. *About CSPC / Transparency Seal. Section 4 Facts and Figures*. Retrieved April 23, 2026 from <https://cspc.edu.ph/exact-page-url>
- [6] Prabhaker Mishra, Chandra M. Pandey, Uttam Singh, and Abhinav Gupta. 2023. Proportional stratified sampling. *Methodological Literature References*.
- [7] Mark Saunders, Philip Lewis, and Adrian Thornhill. 2023. *Research methods for business students*. Pearson Education.
- [8] Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers, and Keying Ye. 2021. *Probability and Statistics for Engineers and Scientists* (9th ed.). Pearson.

-
- [9] Thomas P. Vartanian. 2021. Secondary data analysis. Oxford Research Encyclopedia of Social Work. <https://doi.org/10.1093/acrefore/9780197289747.013.11>
- [10] Fernando Almeida. 2022. Document analysis technique in the context of academic research. *Journal of Education and Learning* 11, 3 (2022), 1–6. <https://doi.org/10.5539/jel.v11n3p1>
- [11] David Byrne. 2022. A worked example of Braun and Clarke’s approach to reflexive thematic analysis. *Quality & Quantity* 56 (2022), 1391–1412. <https://doi.org/10.1007/s11135-021-01182-y>
- [12] I. Kusmaryono, D. Wijayanti, and H. R. Maharani. 2022. Number of Response Options, Reliability, Validity, and Potential Bias in the Use of the Likert Scale in Education and Social Science Research: A Literature Review. *International Journal of Educational Methodology* 8, 4 (2022), 625-637.
- [13] Sri Rama Chandra Charan Teja Tadi. 2021. Asynchronous Data Processing with Machine Learning: Progressive Web Applications for Seamless Offline Functionality. *International Journal of Science and Research (IJSR)* 10, 11 (2021), 661–666. <https://www.ijsr.net>
- [14] Doaa Abdelfattah, Hesham A. Hassan, and Fatma A. Omara. 2022. A novel role-mapping algorithm for enhancing highly collaborative access control system. *Distributed and Parallel Databases* 40, 3 (2022), 521–558. <https://doi.org/10.1007/s10619-022-07407-9>
- [15] Agariadne D. Samala, Fadhli Ranuharja, Budi R. Fajri, Yullys Indarta, and Wawan Agustiarimi. 2022. ViCT-Virtual Campus Tour Environment with Spherical Panorama: A Preliminary Exploration. *International Journal of Interactive Mobile Technologies (iJIM)* 16, 16 (2022), 136–149. <https://doi.org/10.3991/ijim.v16i16.32889>
- [16] Ronald B. Fernandez, Marnelli Leynes, Mark J. S. Almario, Chanille Aputan, Florence J. T. Iloseo, and James R. P. Manalang. 2025. Smart campus directory: AI-powered navigation system deployment for Universidad de Manila. *International Journal of Engineering Technology Research Management* 9, 10 (2025), 13–20. <https://ijetrm.com/>

-
- [17] Hongjian Pu. 2025. Creating Agile Development Practices in Virtual Workspaces: Exploring the Potential of VR Technologies. In Proceedings of the 2024 International Conference on Virtual Reality Technology (ICVRT 2024), 17–22. <https://doi.org/10.1145/3711496.3711499>
- [18] Sabina C. Necula. 2024. Exploring The Model-View-Controller (MVC) Architecture: A Broad Analysis of Market and Technological Applications. Preprints.org. <https://doi.org/10.20944/preprints202404.1860.v1>
- [19] Simon Anderer, Thomas Kempter, Bernhard Scheuermann, and Sanaz Mostaghim. 2023. Dynamic Optimization of Role Concepts for Role-Based Access Control Using Evolutionary Algorithms. SN Computer Science 4, 416 (2023), 1–17. <https://doi.org/10.1007/s42979-023-01805-1>
- [20] Vinutha M. S. and Prajwal M. R. 2023. A Survey on Search Engine Optimization-Types, Techniques and Factors. International Journal of All Research Education and Scientific Methods (IJARESM) 11, 8 (2023), 2455–6211. <http://www.ijaresm.com>
- [21] Amr Elshahed, Majid K. B. M. Ali, Ahmad S. A. Mohamed, Farah A. B. Abdullah, and Lee J. Aun. 2025. Efficient Pathfinding on Grid Maps: Comparative Analysis of Classical Algorithms and Incremental Line Search. IEEE Access 13 (2025), 98484. <https://doi.org/10.1109/ACCESS.2025.3575168>
- [22] William Stallings. 2023. Cryptography and Network Security: Principles and Practice (8th ed.). Pearson.
- [23] Neda Farhadighalati, Luis A. Estrada-Jimenez, Shirin Nikghadam-Hojjati, and Jose Barata. 2025. A Systematic Review of Access Control Models: Background, Existing Research, and Challenges. IEEE Access 13 (2025), 17777-17806. <https://doi.org/10.1109/ACCESS.2025.3533145>

-
- [24] OWASP Foundation. 2021. *OWASP Top 10:2021 - The Ten Most Critical Web Application Security Risks*. <https://owasp.org/Top10/>
- [25] ISO/IEC. 2011. Systems and software engineering Systems and software Quality Requirements and Evaluation (SQuaRE) System and software quality models. ISO/IEC 25010:2011. International Organization for Standardization.